

Functional Implications of the Exon 9 Splice Insert in GluK1 Kainate Receptors

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Abstract

Kainate receptors are key modulators of synaptic transmission and plasticity in the central nervous system. Different kainate receptor isoforms with distinct spatiotemporal expressions have been identified in the brain. The GluK1-1 splice variant receptors, which are abundant in the adult brain, have an extra fifteen amino acids inserted in the amino-terminal domain (ATD) of the receptor resulting from alternative splicing of exon 9. However, the functional implications of this post-transcriptional modification are not yet clear. We employed a multi-pronged approach using cryogenic electron microscopy, electrophysiology, and other biophysical and biochemical tools to understand the structural and functional impact of this splice insert in the extracellular domain of GluK1 receptors. Our study reveals that the splice insert alters the key gating properties of GluK1 receptors and their modulation by the cognate auxiliary Neuropilin and tolloid-like (Neto) proteins 1 and 2. Mutational analysis identified the role of crucial splice residues that influence receptor properties and their modulation. Furthermore, the cryoEM structure of the variant shows that the presence of exon 9 in GluK1 does not affect the receptor architecture or domain arrangement in the desensitized state. Our study thus provides the first detailed structural and functional characterization of GluK1-1a receptors, highlighting the role of the splice insert in modulating receptor properties and their modulation.

eLife assessment

This **important** study shows that a splice variant of the kainate receptor Glu1-1a that inserts 15 amino acids in the extracellular N-terminal region substantially changes the channel's desensitization properties, the sensitivity to glutamate and kainate, and the effects of modulatory Neto proteins. In the revised paper the authors have clarified several points raised by reviewers but the structural portion of the study has not been improved and consequently, more data are needed to determine the molecular mechanism by which the insert changes the functional profile of the channel. Even so, these **solid** findings advance our understanding of splice variants among glutamate receptors and will be of interest to neuro- and cell-biologists and biophysicists in the field.

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Introduction

Kainate receptors (KARs), a subfamily of ionotropic glutamate receptors (iGluRs), are required in the vertebrate brain for postsynaptic neurotransmission and presynaptic regulation of transmitter release (Erreger et al., 2004 [↗](#); Huettner, 2003 [↗](#); Lerma, 2003 [↗](#); Pinheiro and Mulle, 2006 [↗](#)). They are known to mediate characteristic small-amplitude excitatory postsynaptic currents (EPSC) with slow kinetics in the hippocampal regions of the central nervous system compared to their counterparts, α -amino-3-hydroxy-5-methyl-4-isoxazole propionic acid (AMPA) and N-methyl-D-aspartate (NMDA) receptors (Castillo et al., 1997 [↗](#); Lerma et al., 2001 [↗](#); Traynelis et al., 2010 [↗](#)). Furthermore, they play a vital role in the maturation of neuronal circuits during development by interacting with G proteins (Lerma, 2003 [↗](#); Lerma and Marques, 2013 [↗](#)). Any functional defect predisposes the brain to various disorders, such as autism, epilepsy, schizophrenia, and neuropathic pain (Lerma and Marques, 2013 [↗](#); Bowie, 2008 [↗](#); Valbuena and Lerma, 2021 [↗](#)).

Kainate receptors are composed of subunits from two gene families: low kainate affinity and high kainate affinity. The first gene family includes GluK1-GluK3, which can form both homomeric and heteromeric receptors with subunits from the same family or from the high-kainate affinity family. On the other hand, GluK4-GluK5 are high-affinity subunits that must assemble with low-affinity subunits to create functional receptors. The assembly of different subunits results in the formation of various receptor configurations, which contribute to the wide range of functional properties observed in kainate receptors.

The functional repertoire of kainate receptors is further enhanced by RNA editing (Barbon and Barlati, 2011 [↗](#); Seeburg, 2002 [↗](#)) and alternative splicing (Jaskolski et al., 2004 [↗](#)). In particular, GluK1-containing KARs mainly present in the hippocampus, cortical interneurons, Purkinje cells, and sensory neurons undergo alternative splicing and RNA editing (Bernard and Khrestchatsky, 1994 [↗](#)). Alternative splicing of the C-terminal domain that produces four isoforms, GluK1-a (shortest C-terminal), GluK1-b, GluK1-c, and GluK1-d (Bettler et al., 1990 [↗](#); Gregor et al., 1993 [↗](#); Pinheiro and Mulle, 2006 [↗](#); Sommer et al., 1992 [↗](#)) has been studied (Ren et al., 2003 [↗](#)). However, the functional impact of the mRNA splicing of the N-terminal domain (exon 9) resulting in isoforms GluK1-1 and GluK1-2, with GluK1-1 containing an additional 15 amino acids in the amino-terminal domain (ATD) is not understood. Interestingly, the splice junctions for GluK1-1 were reported to be more frequent than those for GluK1-2, and both variants showed similar expression in different species (Herbrechter et al., 2021 [↗](#)). This 15-amino acid insertion in the amino-terminal domain (ATD) is exclusive to the GluK1 subunit and may impart unique properties to KARs containing this variant. Therefore, investigations of functional differences between GluK1 receptors with and without the 15 amino acid splice insert are necessary but missing in the field. Furthermore, the impact of the N-terminal splice insert in GluK1-1 receptors on its modulation by cognate auxiliary subunits (Fisher, 2015 [↗](#); He et al., 2021a [↗](#); Sheng et al., 2015 [↗](#); Vinnakota et al., 2021 [↗](#)), neuropilin and tolloid-like (Neto) proteins is also unknown.

Therefore, we conducted structure-function studies to understand the mechanisms of GluK1-1a receptor function and the effects of the ATD splice on receptor assembly, stability, and modulation by Neto proteins. Whole-cell and excised outside-out patch-clamp-based functional assays were performed to investigate the differences between the ATD splice variants GluK1-1a (exon 9) and GluK1-2a (lacking exon 9) and their modulation by Neto proteins. Furthermore, mutational analysis was performed to identify important splice residues that affect receptor functions and their modulation by Neto proteins. Additionally, we expressed and purified rat GluK1-1a from HEK293 GnTI-cells and determined the single-particle cryo-EM structures of the receptors trapped in the desensitized state to understand the effect of splice on receptor architecture.

Results

Spatiotemporal expression pattern of GluK1-1 in the brain

Kainate receptor subunits are differentially and spatially regulated in vertebrates. We analyzed publicly available human transcriptomics data to determine whether the exon encoding the GluK1 ATD splice (exon 9) was differentially expressed in the human brain. RNASeq data analysis of *GRIK1* (Ensembl ID: ENSG00000171189) collected from the BrainSpan ATLAS indicated that exon 9 is present in multiple brain areas that also show prominent GluK1 expression. *GRIK1* exon 9 expression varies significantly across brain regions and developmental stages, with dynamic patterns indicating its crucial role in brain development. High expression is noted during critical periods such as early embryonic, postnatal, and childhood stages, particularly in regions like the ventricular zone (VZ), cortical plate (CP), hippocampus (HIP), amygdala (AMY), striatum (STR), and caudate nucleus (CN) (**Figure 1** [↗](#); **Figure 1-figure supplement 1**). These patterns suggest *GRIK1* exon 9's importance in the functional maturation of these brain regions. For example, in the cerebellar cortex, *GRIK1* exon 9 expression peaks during early development and childhood and stabilizes into adulthood, underscoring its role in cortical development and function. (**Figure 1-figure supplement 1**). The significant variability in exon 9 expression across different brain regions, along with its potential role in brain development and functional maturation, motivated us to investigate the underlying molecular mechanisms. This prompted us to perform a detailed structure-function analysis of the GluK1-1a splice variant.

The ATD splice imparts functional diversity to GluK1 receptors

Previous studies on functional analysis of GluK1 receptors have primarily focused on GluK1-2 isoform that lacks the ATD splice insert. Therefore, we conducted an extensive electrophysiological investigation to assay the functional differences between GluK1-1a and GluK1-2a. The only difference between these two variants is the presence of 15 amino acids (KASGEVSKHLYKVWK) insertion in the R2 subdomain (lower lobe) of the GluK1-1a ATD. We evaluated multiple gating properties of the two variants by whole-cell patch-clamp electrophysiology. Interestingly, we observed that GluK1-1a receptors desensitize significantly slowly when compared to GluK1-2a (T_{Des} , GluK1-1a: 5.21 ± 0.50 ms; GluK1-2a: 3.55 ± 0.23 ms $^{***}P=0.0092$) on prolonged treatment with 10 mM glutamate (**Figure 2A** [↗](#); **Table 1** [↗](#)). We also tested receptor desensitization by applying saturating concentrations of kainate (1mM). Since the receptors displayed prolonged desensitizing kainate currents, instead of the rate (T_{Des}), we calculated the % desensitization values at 1 s for comparison. We observed that the % desensitization with kainate was distinct for both ATD splice variants with significantly slower desensitization in the variant with the splice insert (GluK1-1a: 72.06 ± 2.33 %, GluK1-2a: 93.2 ± 0.55 $^{***}P=0.0006$) consistent with the glutamate evoked currents (**Figure 2B** [↗](#); **Table 1** [↗](#)).

Next, we evaluated glutamate sensitivity for the two variants. Dose-response experiments (glutamate, GluK1-1a: 0.1-2 mM, and GluK1-2a: 0.01-3 mM) showed a significantly lower potency of glutamate for GluK1-1a variant compared to the non-spliced form (EC_{50} glutamate, GluK1-1a: 379.3 ± 52 μ M, GluK1-2a: 187.7 ± 33 μ M $^{*}P=0.0129$ (**Figure 2C** [↗](#); **Table 1** [↗](#)). However, for high-affinity agonist kainate, the dose-response curves for both variants were similar, likely indicating differences in the stability of glutamate *versus* the kainate-bound states in GluK1-1a and GluK1-2a (**Figure 2D** [↗](#); **Figure 2-figure supplement 1**). Thus, the potency of kainate (1mM) *versus* glutamate (10 mM) (I_K/I_G ratio) is significantly higher for GluK1-1a compared to GluK1-2a (GluK1-1a: 1.51 ± 0.13 , GluK1-2a: 0.56 ± 0.4 $^{****}P<0.0001$) (**Figure 2D** [↗](#); **Table 1** [↗](#)).

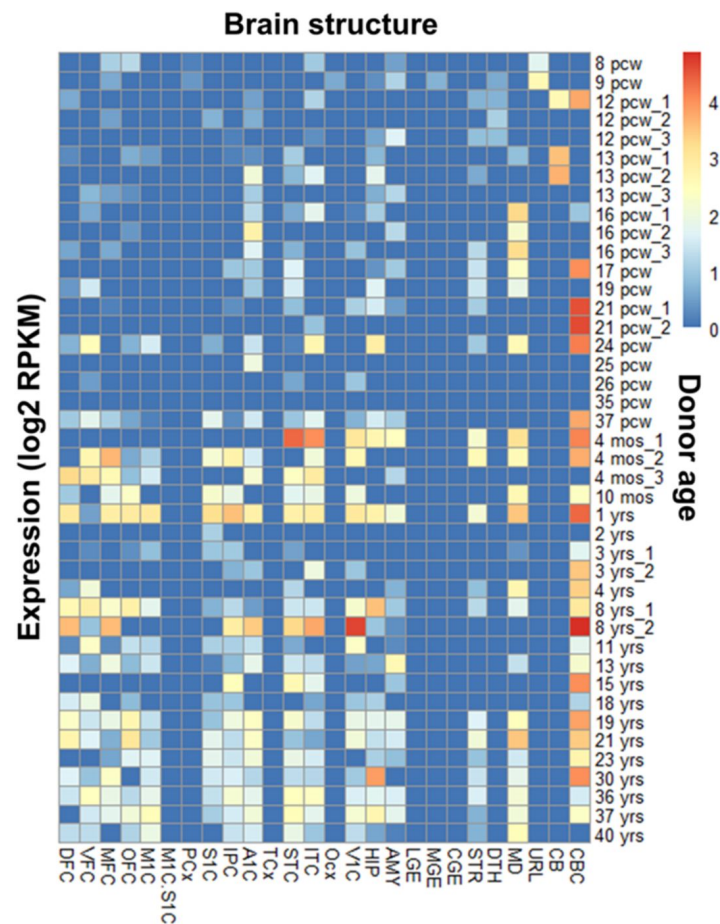


Figure 1.

RNA-Seq analysis (BrainSpan atlas) demonstrates an abundance of *GRIK1* exon 9 in the human brain. The heat map shows the presence of exon 9 (45 bp; ENSE00001313812) that codes for GluK1 ATD splice in different brain regions from the embryonic to adult stage. Exon 9 expression coincides with well-studied areas for GRIK1 gene like cerebellar cortex, visual cortex, etc. Various regions of the brain and the donor age are represented on the x-axis and y-axis respectively. The donor age has been abbreviated as pcw (post-conception weeks), mos (months), and yrs (years). The regions of the human brain are abbreviated as: DFC (dorsolateral prefrontal cortex), VFC (ventrolateral prefrontal cortex), MFC (anterior [rostral] cingulate [medial prefrontal] cortex), OFC (orbital frontal cortex), M1C (primary motor cortex area M1, area 4), M1C.S1C (primary motor-sensory cortex [samples]), PCx (parietal neocortex), S1C (primary somatosensory cortex (area S1, areas 3,1,2), IPC (posteroventral [inferior] parietal cortex), A1C (primary auditory cortex core), A1C (primary auditory cortex [core]), TCx (temporal neocortex), STC (posterior [caudal] superior temporal cortex, area 22c), ITC (inferolateral temporal cortex (area TEv, area 20), OCx (occipital neocortex), V1C (primary visual cortex (striate cortex, area V1/17), HIP (hippocampus), AMY (amygdaloid complex), LGE (lateral ganglionic eminence), MGE (medial ganglionic eminence), CGE (caudal ganglionic eminence), STR (striatum), DTH (dorsal thalamus), MD (mediodorsal nucleus of thalamus), URL (upper [rostral] rhombic lip), CB (cerebellum) and CBC (cerebellar cortex). Blue and red color indicate zero and maximum expression respectively.

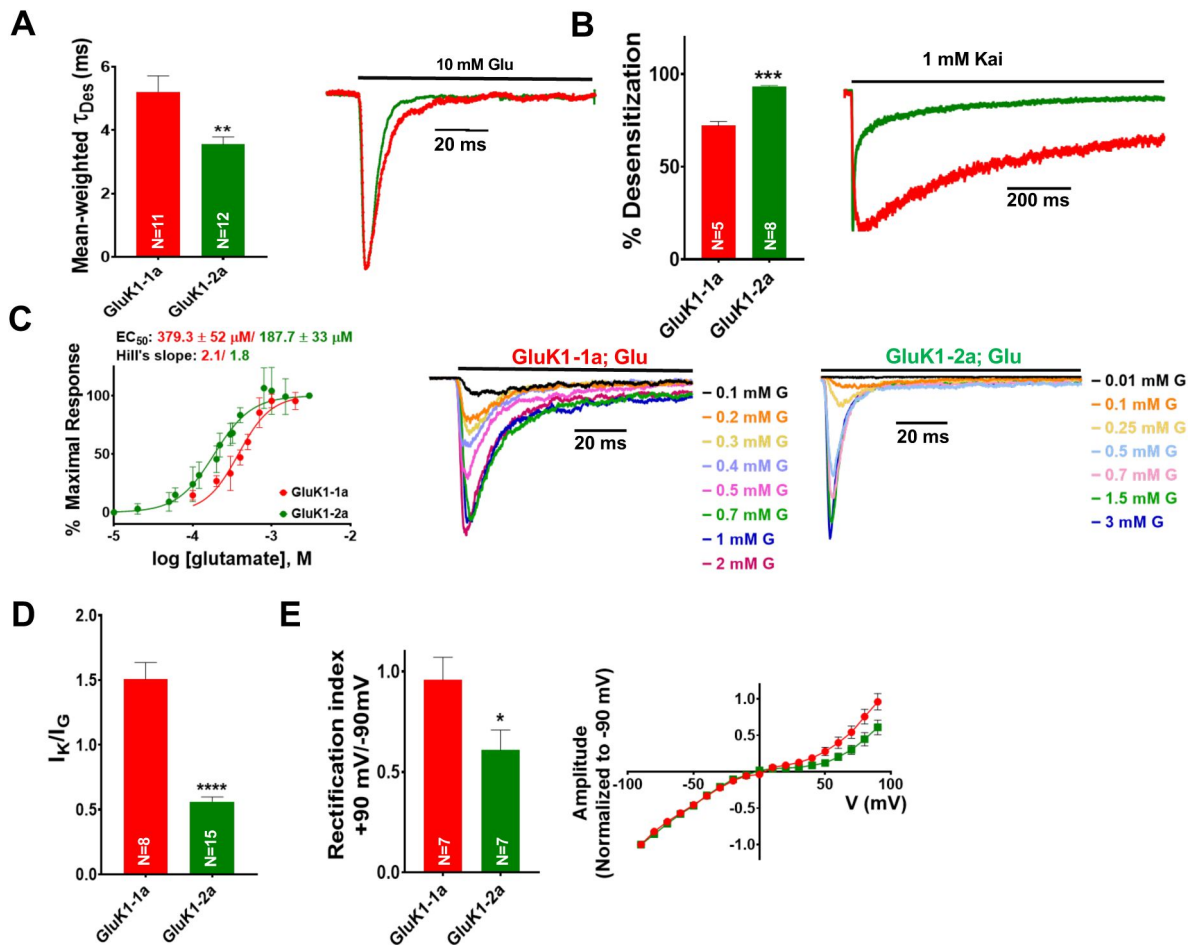


Figure 2.

ATD splice insert affects the gating properties of the GluK1-1a homomeric receptors. **(A)** Displays mean-weighted Tau (τ_{Des}) values for GluK1-1a wild-type (red) and GluK1-2a wild-type (green) in the presence of glutamate. The inset shows representative normalized traces for GluK1-1a and GluK1-2a with 10 mM glutamate. **(B)** Displays the percent desensitization values calculated at 1 s for GluK1-1a wild-type (red) and GluK1-2a wild-type (green) in the presence of kainate. Representative normalized traces for GluK1-1a and GluK1-2a with 1 mM kainate are shown. **(C)** Demonstrates glutamate dose-response curves for GluK1-1a and GluK1-2a. Representative aligned traces for both receptors at various glutamate concentrations are shown. The kainate dose responses for the splice variants are shown in **Figure 2-supplement figure 1**. **(D)** The ratio of currents evoked by kainate and glutamate is plotted for GluK1-1a and GluK1-2a. **(E)** The ratio of currents evoked by the application of 10 mM glutamate at +90 mV and -90 mV for the GluK1-1a and GluK1-2a receptors is shown. Representative IV plots are depicted for GluK1-1a and GluK1-2a for the entire voltage ramp (-90 to +90 mV). **(F)** Error bars indicate mean $\pm$ SEM, N in each bar represents the number of cells used for analysis, and * indicates the significance at a 95% confidence interval.

S. No.	Name of the construct	Desensitization					Total No. of cells tested
		10mM G (T_{Des} in ms)	1 mM K (%)	I_K/I_G	Recovery (T_{Rec} in s)	Rectification (+90 mV/-90 mV)	
1	<i>GluK1-1a_{EM}</i>	1.94±0.13 N=8 (****P<0.0001)	-	-	2.63±0.04 N=3 (P= 0.3501))	-	23
2	<i>GluK1-1a</i>	5.21±0.50 N=11	72.06±2.33 N=5	1.51±0.13 N=8	3.53±0.81 N=4	0.96±0.11 N=7	19
	+ <i>Neto1</i>	3.56±0.22 N=9 (**P=0.0090)	49.61±5.23 N=7 (**P=0.0075)	1.25±0.04 N=8 (P=0.1500)	0.68±0.07 N=8 (*P=0.0390)	1.16±0.09 N=7 (P=0.1880)	11
	+ <i>Neto2</i>	69.62±9.98 N=6 (**P=0.0024)	68.96±4.36 N=9 (P=0.7459)	1.00±0.07 N= 13 (**P=0.0096)	8.32±0.81 N=5 (**P=0.0044)	0.80±0.05 N=4 (P=0.2401)	17
3	<i>GluK1-2a</i>	3.55±0.23 N=12 (**P=0.0092)	93.20±0.55 N=8 (***P= 0.0006)	0.56±0.04 N=15 (****P<0.0001)	5.31±0.50 N=7 (P=0.1191)	0.61±0.10 N=7 (*P=0.0385)	21
	+ <i>Neto1</i>	4.32±0.34 N=8 (P=0.1495)	76.99±3.41 N=6 (**P=0.0085)	1.37±0.12 N=10 (****P <0.0001)	1.15±0.12 N=5 (***P=0.0002)	1.14±0.14 N=3 (*P=0.0338)	14
	+ <i>Neto2</i>	21.68±2.64 N=6 (**P=0.0017)	65.98±2.41 N=7 (****P<0.0001)	1.16±0.04 N=8 (****P <0.0001)	7.91±0.71 N=4 (*P=0.0430)	1.37±0.10 N=3 (**P= 0.0022)	18
4	<i>K₃₇₉H₃₇₆-A</i>	3.80±0.33 N=5	71.07±5.27 N=5	1.43±0.07 N=5	1.10±0.22 N=4	0.78±0.05 N=4	9
	+ <i>Neto1</i>	3.38±0.14 N=8	54.64±5.18 N=6	1.76±0.14 N=8	1.15±0.11 N=9	0.64±0.04 N=4	10
	+ <i>Neto2</i>	42.57±11.96 N=4	61.03±5.12 N=5	1.02±0.02 N=5	9.32±0.15 N=3	0.60±0.06 N=5	6
5	<i>K_{375/379/382}-A</i>	Low peak amplitudes (≤40 pA)					15
	+ <i>Neto1</i>	2.82±0.18 N=6	64.13±5.26 N=5	1.42±0.08 N=6	0.74±0.08 N= 6	1.28±0.10 N=6	8
	+ <i>Neto2</i>	Low peak amplitudes (≤40 pA)					27
6	<i>Y₃₇₂V₃₈₀W₃₈₁-A</i>	Low peak amplitudes (≤40 pA)					18
	+ <i>Neto1</i>	3.31±0.16 N=9	71.48±4.66 N=5	1.76±0.21 N=7	1.02±0.07 N=7	0.77±0.14 N=5	10
	+ <i>Neto2</i>	45.79±17.32 N=3	51.82±4.88 N=3	1.19±0.11 N=3	9.45±1.21 N=3	1.03±0.44 N=4	22
7	<i>K₃₇₅-K₃₈₂-8A</i>	Non-functional N=15	Non-functional N=5	-	-	Non-functional N=5	15
	+ <i>Neto1</i>	Non-functional N=11	Non-functional N=4	-	-	Non-functional N=5	11
	+ <i>Neto2</i>	Non-functional N=11	Non-functional N=5	-	-	Non-functional N=6	11
8	<i>K₃₇₂H₃₇₆-E</i>	Low peak amplitudes (≤40 pA)					13
	+ <i>Neto1</i>	3.59±0.29 N=4	13.78±3.16 N=5	0.70±0.19 N=4	0.63±0.10 N=3	0.68±0.19 N=3	23
	+ <i>Neto2</i>	Low peak amplitudes (≤40 pA)					36
9	<i>H₃₇₆-E</i>	5.07±0.42 N=7	29.80±13.21 N=4	0.88±0.05 N=3	1.48±0.18 N=8	0.44±0.07 N=6	19
	+ <i>Neto1</i>	2.69±0.15 N=5	59.24±8.62 N=4	2.36±0.30 N=4	0.66±0.05 N=5	0.80±0.19 N=4	6
	+ <i>Neto2</i>	50.88±6.49 N=4	28.82±6.43 N=5	1.43±0.13 N=4	4.98±0.81 N=3	0.42±0.08 N=8	50
10	<i>K_{375/379/382}H₃₇₆-E</i>	9.62±1.47 N=6 (*P=0.0290)	54.25±8.28 N=3 (*P=0.0396)	1.17±0.28 N=3 (P=0.3733)	4.83±0.31 N=3 (P=0.2774)	0.62±0.14 N=7 (*P=0.0499)	36
	+ <i>Neto1</i>	3.74±0.20 N=8 (P=0.5390)	57.72±12.65 N=3 (P=0.5996)	1.69 ±0.15 N=6 (*P=0.0326)	0.66±0.08 N=8 (P=0.8702)	1.23±0.07 N=6 (P=0.5351)	9
	+ <i>Neto2</i>	65.57±17.04 N=8 (P=0.8412)	26.66±2.05 N=4 (****P<0.0001)	0.86±0.08 N=4 (P=0.2045)	6.57±0.61 N=3 (P=0.3588)	0.72±0.09 N=4 (P=0.6677)	48
11	<i>K_{368/375/379/382}H₃₇₆-E</i>	Low peak amplitudes (≤40 pA)					20
	+ <i>Neto1</i>	4.86±0.67 N=5 (P=0.1242)	10.39±.3.53 N=5	1.15±0.20 N=4 (P=0.6495)	0.21±0.12 N=3	0.59±0.13 N=5 (**P=0.0081)	21
	+ <i>Neto2</i>	Low peak amplitudes (≤40 pA)					22
12	<i>K₃₆₈-E</i>	7.89±1.14 N=3 (*P=0.0334)	46.66±5.88 N=5 (**P=0.0093)	0.81±0.13 N=5 (**P=0.0037)	5.61±0.95 N=3 (*P=0.0417)	a N=7 (****P<0.0001)	32
	+ <i>Neto1</i>	4.37±0.25 N=4 (*P=0.0393)	59.68±8.72 N=3 (P=0.3851)	6.99±0.47 N=5 (***P=0.0002)	0.43±0.05 N=6 (*P=0.0118)	2.84±0.16 N=3 (**P=0.0021)	8
	+ <i>Neto2</i>	79.40±4.44 N=5 (P=0.4012)	33.47±8.98 N=5 (*P=0.0388)	2.13±0.13 N=4 (***P=0.0007)	7.66±0.83 N=3 (P=0.9473)	0.76±0.26 N=8 (P=0.9860)	43

Table 1.

Whole-Cell patch clamp recordings of GluK1-1a, GluK1-1a_{EM}, GluK1-2a, and GluK1-1a mutants in absence or presence of Neto1 (green) or Neto2 (peach).

Errors are reported as SEM. Statistical significance is reported at 95 % CI. P<0.05 (*), P<0.01 (**), P<0.001 (***), P<0.0001 (****) for comparisons between wild-type GluK1-1a receptor with EM construct, GluK1-2a, or various mutants in presence or absence of Neto proteins.

‘a’ denotes the lack of rectification index value due to no conductance observed at positive potentials in the K₃₆₈-E mutant alone.

Further, we investigated the voltage-dependent endogenous polyamine block and found significant differences between the two variants. The presence of splice residues seemed to enhance outward currents at positive potentials in GluK1-1a compared to GluK1-2a (rectification index, +90mV /-90 mV; GluK1-1a = 0.96 ± 0.11 ; GluK1-2a = 0.61 ± 0.10 * $P=0.0385$) without affecting the reversal potential (**Figure 2E**; **Table 1**). How splice residues situated ~92 Å away from the TM domain affect the pore properties is unclear. This property is mainly affected by the TM2 region of the KAR (Bowie and Mayer, 1995). However, a recent report in which the ATD of GluK2 was deleted showed enhanced rectification (Li et al., 2019). Splice residues likely alter pore structure by allosteric mechanisms that have yet to be identified, thereby affecting rectification (Perrais et al., 2009). We also performed outside-out patch recordings to examine GluK1-1a receptors. However, we observed extremely weak electrical currents with low amplitudes when GluK1-1a was expressed in isolation. Consequently, the data we obtained from these recordings did not provide reliable results for curve fitting or thorough analysis.

ATD splice insert impacts GluK1 receptor modulation by Neto proteins

Neto 1 and Neto 2 proteins significantly influence the surface expression, synaptic localization, and functional properties of the GluK1-2a receptors (Copits et al., 2011; Palacios-Filardo et al., 2016; Sheng et al., 2015). It has been demonstrated that desensitization of GluK1-2a is accelerated by Neto1 but delayed by Neto2 (Sheng et al., 2015; Copits et al., 2011; Palacios-Filardo et al., 2016). Hence, to understand the influence of these KAR auxiliary proteins on GluK1-1a, we performed an electrophysiological analysis of GluK1-1a and GluK1-2a receptors co-expressed with either Neto1 or Neto2. Co-expression of Neto1 hastened desensitization of GluK1-1a significantly but not of GluK1-2a at saturating glutamate concentrations (T_{Des} , GluK1-1a +Neto1: 3.56 ± 0.22 ms ** $P=0.0090$; GluK1-2a +Neto1: 4.32 ± 0.34 P=0.1495). On the other hand, Neto2 led to a ~ 13.4-fold decrease in the desensitization rate of GluK1-1a (T_{Des} , GluK1-1a: 5.21 ± 0.50 ms, +Neto2: 69.62 ± 9.98 ms ** $P=0.0024$) while the desensitization rate of GluK1-2a was decreased only by ~ 6.1-fold (T_{Des} , GluK1-2a: 3.55 ± 0.23 ms, +Neto2: 21.68 ± 2.64 ms ** $P=0.0017$). Thus, while Neto1 accelerated the desensitization of GluK1-1a, Neto2 significantly slowed it at saturating glutamate concentrations. The rate of desensitization for GluK1-1a was approximately 3.2 times slower compared to GluK1-2a (** $P=0.0009$) when coexpressed with Neto2 (**Figure 3A**; **Table 1**). Thus, the presence of splice insert leads to differential modulation of GluK1 desensitization by Neto proteins.

Moreover, Neto1 also enhanced the recovery from desensitization for both the variants ($T_{Recovery}$, GluK1-1a: 3.53 ± 0.81 s, +Neto1: 0.68 ± 0.07 s * $P=0.0390$; GluK1-2a: 5.31 ± 0.50 s +Neto1: 1.15 ± 0.12 s *** $P=0.0002$). GluK1-1a recovers ~ 1.7 times faster than GluK1-2a (* $P=0.0125$) when co-expressed with Neto1. Neto 2, on the other hand, slowed recovery for both variants to a similar extent ($T_{Recovery}$, GluK1-1a: 3.53 ± 0.81 s, +Neto2: 8.32 ± 0.81 s ** $P=0.0044$; GluK1-2a: 5.31 ± 0.50 s, +Neto2: 7.91 ± 0.71 s * $P=0.0430$) and did not show differential modulation (**Figure 3B**; **Table 1**). In addition, both Neto1 and Neto2 increased the potency of glutamate for GluK1-1a by 9.7-fold (39 ± 10 μM) and 11.2-fold (34 ± 8 μM), respectively (**Figure 3C**; **Figure 3-figure supplement 1A**). This is similar to the effect observed in GluK1-2a receptors whereby the glutamate EC_{50} was shown to increase by Neto proteins [Neto1: 34-fold and Neto2: 7.5-fold (Palacios-Filardo et al., 2016) and Neto1/2: 30-10X (Fisher, 2015)].

Since we observed a significant increase in the I_K/I_G ratios in GluK1 due to the presence of splice insert, we next aimed to determine the influence of co-expressing Neto proteins on this parameter. Interestingly, while Neto 1 and Neto2 reduced the I_K/I_G ratios in GluK1-1a (GluK1-1a: 1.51 ± 0.13 , +Neto1: 1.25 ± 0.04 P=0.1500, +Neto2: 1.0 ± 0.07 ** $P=0.0090$), they increased it for the non-spliced variant (GluK1-2a: 0.56 ± 0.04 , +Neto1: 1.37 ± 0.12 **** $P<0.0001$, +Neto2: 1.16 ± 0.04 **** $P<0.0001$)

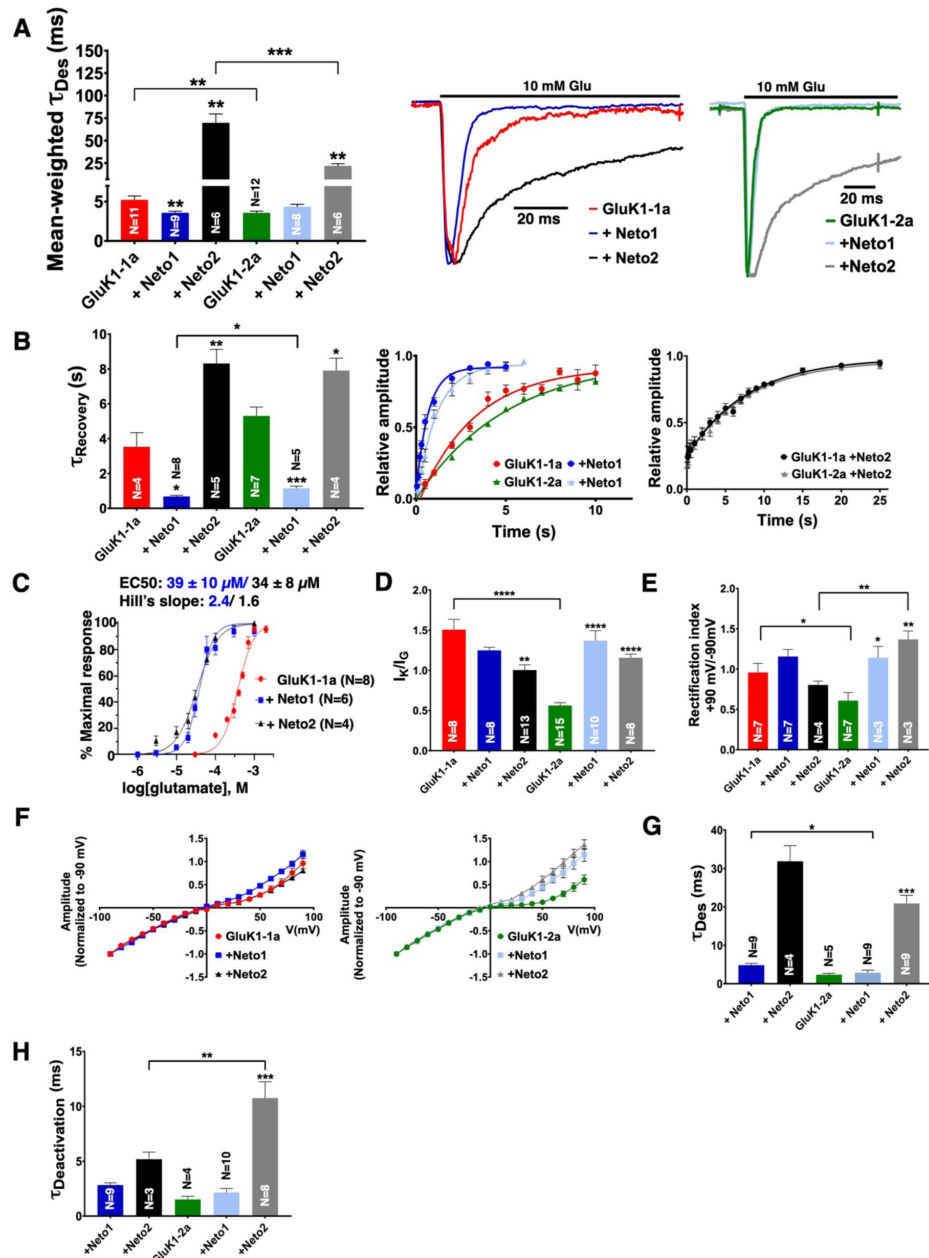


Figure 3.

ATD splice affects the functional modulation of GluK1 kainate receptors by Neto proteins. **(A)** Shows mean-weighted Tau (τ_{Des}) values calculated at 100 ms for GluK1-1a (red) and GluK1-2a (green), respectively, with full-length Neto1 (blue/light blue) or Neto2 (black/gray), in the presence of glutamate. Representative normalized traces are shown for 100 ms application of 10 mM glutamate for HEK293 cells co-expressing GluK1-1a or GluK1-2a with Neto1 and Neto2. **(B)** Shows Tau ($\tau_{Recovery}$) values plotted for GluK1-1a and GluK1-2a, respectively, with full-length Neto1 or Neto2. Relative amplitude graphs for each receptor in the absence or presence of Neto proteins are also depicted. **(C)** Demonstrates the glutamate dose-response curves for GluK1-1a with Neto proteins. **(D)** Indicates the ratio of peak amplitudes evoked in the presence of 1 mM kainate and 10 mM glutamate for GluK1-1a or GluK1-2a with or without Neto proteins. **(E)** The ratio of currents evoked by the application of 10 mM glutamate at +90 mV and -90 mV for the receptors in the absence or presence of Neto proteins is shown. **(F)** Shows representative IV plots for GluK1-1a and GluK1-2a for the receptor alone versus with Neto proteins, respectively. Panels **G** and **H** show data recorded from outside-out pulled-patches. **(G)** Displays desensitization kinetics for GluK1-1a (red) and GluK1-2a (green) with or without Neto proteins, respectively. **(H)** Shows deactivation kinetics at 1 ms for GluK1-1a (red) and GluK1-2a (green) with or without Neto proteins. Error bars indicate mean $\pm$ SEM, N in each bar represents the number of cells used for analysis, and * indicates the significance at a 95% confidence interval.

(**Figure 3D** [↗](#); **Figure 3-figure supplement 1B**; **Table 1** [↗](#)) (Zhang et al., 2009 [↗](#)) highlighting the effect of splice residues. Interestingly, this differential modulation of the two variants by Neto proteins resulted in comparable I_K/I_G ratios.

Next, we investigated the effects of Neto1 and Neto2 on the voltage-dependent endogenous polyamine block since the presence of splice insert had enhanced the outward rectification of GluK1. We observed that both Neto1 and Neto2 significantly enhanced the outward rectification of GluK1-2a (GluK1-2a: 0.61 ± 0.10 , +Neto1: 1.14 ± 0.14 * $P=0.0338$, +Neto2: 1.37 ± 0.10 ** $P=0.0022$). However, they did not significantly increase it for GluK1-1a (GluK1-1a: 0.96 ± 0.11 , +Neto1: 1.16 ± 0.09 $P=0.1880$, +Neto2: 0.80 ± 0.05 $P=0.2401$) and did not show differential modulation (**Figure 3E** [↗](#) & **3F**; **Table 1** [↗](#)).

We also calculated desensitization and deactivation kinetics in excised outside-out patches. However, GluK1-1a receptors, when expressed alone, exhibited low peak amplitudes in outside-out recordings that prevented reliable calculation of gating kinetics. Hence, we only compared the properties of receptors co-expressed with Neto proteins, and the results were consistent with those obtained from whole-cell recordings (**Figure 3G-I** [↗](#), **Table 2** [↗](#)). Neto 2 slowed down desensitization of GluK1-1a by ~ 1.5 times compared to GluK1-2a (T_{Des} , GluK1-1a +Neto2: 31.89 ± 4.08 ms; GluK1-2a +Neto2: 20.91 ± 2.11 ; $P=0.0665$) (**Figure 3G** [↗](#), **Table 2** [↗](#)). The desensitization rates for GluK1-1a and GluK1-2a receptors co-expressed with Neto1 was also significantly altered (T_{Des} , GluK1-1a +Neto1: 4.83 ± 0.46 ms; GluK1-2a +Neto1: 2.84 ± 0.35 ; * $P=0.0310$) (**Figure 3G** [↗](#), **Table 2** [↗](#)). Faster solution exchange times in excised patch recordings also allowed us to measure deactivation kinetics using a 1 ms application of 10 mM glutamate. Surprisingly, unlike desensitization, for receptors coexpressed with Neto2, the deactivation rate of GluK1-1a is significantly faster compared to that of GluK1-2a (T_{Dea} , GluK1-1a +Neto2: 5.18 ± 0.65 ms; GluK1-2a +Neto2: 10.74 ± 1.48 ; ** $P=0.0077$). In contrast, Neto1 did not significantly alter deactivation kinetics of both GluK1 variant receptors (T_{Dea} , GluK1-1a +Neto1: 2.83 ± 0.20 ms; GluK1-2a +Neto1: 2.14 ± 0.37 ; $P=0.2086$) (**Figure 3H** [↗](#), **Table 2** [↗](#)). Thus, both whole-cell and excised patch recordings confirm the unique functional properties and differential modulation by Neto proteins due to the presence of fifteen amino acid insert in GluK1-1a receptors.

Mutations in GluK1-1a splice insert alter channel properties

Since our electrophysiological analysis showed functional differences between the two splice variants and their modulation by Neto proteins, we created receptors with mutated splice residues and conducted functional assays to identify the key residues. The splice insert (KASGEVSKHLYKVWK) is dominated by positively charged residues and contains four lysins and one histidine. We hypothesized that these charged residues might affect the interactions at the ATD-LBD interface and influence receptor functions. To investigate this, we prepared charge-reversal (K/H to E) and charge-neutral (K/H to A) mutants and carried out functional assays (**Figure 4-table supplement 1**). Our cell surface biotinylation assay showed that all mutants (glutamate or alanine) reached the cell surface efficiently (data not shown). However, the charge-neutral mutants (K/H to A) gave either very low peak amplitudes (<40 pA) or were not functional and hence, were not included in the study. The charge reversal mutants K_{368} -E and $K_{375/379/382}H_{376}$ -E revealed fascinating insights into the role of splice residues in altering GluK1 receptor properties (**Figure 4A** [↗](#)). Interestingly, both mutants, K_{368} -E and $K_{375/379/382}H_{376}$ -E exhibited a significantly slower rate of glutamate-evoked desensitization compared to wild-type GluK1-1a (T_{Des} , GluK1-1a: 5.21 ± 0.50 ms; K_{368} -E: 7.89 ± 1.14 ms * $P=0.0334$; $K_{375/379/382}H_{376}$ -E: 9.61 ± 1.47 * $P=0.0290$) (**Figure 4B** [↗](#); **Table 1** [↗](#)). We also observed a significant delay in the recovery from the desensitized state for K_{368} -E mutant (K_{368} : 5.61 ± 0.95 s * $P=0.0417$) compared to wild-type GluK1-1a. In addition, the $K_{375/379/382}H_{376}$ -E mutant also exhibited a slowdown in the recovery, though not significant. ($K_{375/379/382}H_{376}$ -E: 4.83 ± 0.31 s $P=0.2774$) (**Figure 4C** [↗](#); **Table 1** [↗](#)). Our investigations of glutamate- and kainate-evoked responses for wild-type and mutant receptors considering their peak amplitudes (I_K/I_G) revealed a significant decrease for the K_{368} -E mutant

Name of the construct	Deactivation		Desensitization		Total No. of cells tested
	T_{Deact} (ms)	Rise Time (ms)	T_{Des} (ms)	Rise Time (ms)	
GluK1-1a	Low peak amplitude	Low peak amplitude	Low peak amplitude	Low peak amplitude	10
+ Neto1	2.83± 0.2 (N = 9)	0.75 ± 0.04 (N = 9) (P= 0.7615)	4.83 ± 0.46 (N = 9)	0.71 ± 0.04 (N = 9)	9
+ Neto2	5.18 ± 0.65 (N = 3)	1.5 ± 0.24 (N = 3) (P= 0.0627)	31.89 ± 4.08 (N = 4)	1.87 ± 0.06 (N = 4)	4
GluK1-2a	1.51±0.28 (N = 4)	0.58 ± 0.06 (N = 4)	2.34 ± 0.35 (N = 5)	0.72 ± 0.07 (N = 5)	9
+ Neto1	2.14±0.37 (N = 10) (P=0.2086)	0.75 ± 0.06 (N = 10) (P=0.0731)	2.84 ± 0.69 (N = 9) (P=0.5312/ *P=0.0310)	0.69 ± 0.07 (N = 9)	12
+ Neto2	10.74 ±1.48 (N = 8) (***P=0.0004/ **P=0.0077)	1.47 ± 0.23 (N = 8) (P=0.062)	20.91 ± 2.11 (N = 9) (***P=0.0003/ P=0.0665)	1.42 ± 0.25 (N = 9)	9

Table 2.

Excised patch outside-out electrophysiology of GluK1-1a and GluK1-2a in absence or presence of Neto1 (green) or Neto2 (peach).

Errors are reported as SEM. Statistical significance is reported at 95 % CI. P<0.05 (*), P<0.01 (**), P<0.001 (***), P<0.0001 (****) for comparisons between GluK1-1a and GluK1-2a receptors in presence of Neto proteins. Red p-values are statistical significance at 95% for comparison between GluK1-1a and GluK1-2a with either of the Neto proteins.

(GluK1-1a: 1.51 ± 0.13 ; K₃₆₈-E: 0.81 ± 0.13 **P=0.0037) and a reduction was observed for K_{375/379/382}H₃₇₆-E receptors (1.17 ± 0.28 P=0.3733) compared to wild-type (**Figure 4D** [↗](#); **Table 1** [↗](#)). These observations are reciprocal to the effect of splice insert compared to the non-spliced form, indicating the importance of these residues in influencing receptor desensitization and recovery. A similar trend reversal was also observed for the measurements of the rectification index for these mutants at positive and negative potentials (+90 mV and -90 mV). The rectification index was significantly reduced in the case of mutant K_{375/379/382}H₃₇₆-E (K_{375/379/382}H₃₇₆-E: 0.62 ± 0.14 *P=0.0499). Surprisingly, no outward rectification was observed for the K₃₆₈-E mutant, and further investigation is needed to fully understand the reasons for the same (**Figure 4E** [↗](#); **Figure 5-figure supplement 2C**; **Table 1** [↗](#)).

GluK1-1a splice residues K₃₆₈, K₃₇₅, H₃₇₆, K₃₇₉ and K₃₈₂ influence receptor modulation by Neto proteins

Next, we investigated the effects of splice mutants on receptor modulation by Neto proteins. To test whether mutations in splice residues could disrupt interactions with Neto proteins, we performed receptor pull-downs using an antibody against the His-tag of the receptor. Our results showed that the mutant receptors could efficiently pull down Neto1 (detected using the Neto1 antibody) and Neto2-EGFP (detected using the GFP antibody) (**Figure 5-figure supplement 1**), suggesting that mutants don't altogether abolish GluK1-1a and Neto interactions.

Furthermore, we conducted electrophysiology experiments to investigate whether coexpression of Neto proteins can restore functionality and influence the functions of splice mutants. We observed that mutant K₃₆₈-E desensitizes significantly slower in the presence of Neto1 while the mutants K_{375/379/382}H₃₇₆-E and K_{368/375/379/382}H₃₇₆-E do not exhibit any significant deviation (T_{Des} , GluK1-1a +Neto1: 3.56 ± 0.22 ms; K₃₆₈-E +Neto1: 4.37 ± 0.25 ms *P=0.0393; K_{375/379/382}H₃₇₆-E +Neto1: 3.74 ± 0.20 ms P=0.5390; K_{368/375/379/382}H₃₇₆-E +Neto1: 4.86 ± 0.67 ms P=0.1242) (**Figure 5A** [↗](#); **Figure 5-figure supplement 2A**; **Table 1** [↗](#)). This observation suggests that some splice residues influence GluK1-1a modulation by Neto1, and more mutational, functional, and structural studies on this interaction are necessary. On the other hand, Neto2 does not significantly affect the desensitization of these mutant receptors compared to wild-type-Neto2 (**Figure 5A** [↗](#); **Figure 5-figure supplement 2A**; **Table 1** [↗](#)).

Similarly, K₃₆₈-E and K_{368/375/379/382}H₃₇₆-E mutants recover significantly faster from the desensitized state when coexpressed with Neto1. On the other hand, the recovery from the desensitized state for mutant receptors is not significantly affected by Neto2. Thus, while Neto1 seems to affect the mutant receptor recovery from the desensitized state, Neto 2 doesn't show significant differences compared to wild-type receptors, again highlighting the differential modulation of GluK1-1a receptors by Neto proteins (**Figure 5B** [↗](#); **Table 1** [↗](#)).

Furthermore, to determine whether the agonist efficacy of mutant receptors changed in the presence of Neto proteins, I_K/I_G ratios were measured. Neto1 increased the agonist efficacy for the K₃₆₈-E and K_{375/379/382}H₃₇₆-E mutants but not for K_{368/375/379/382}H₃₇₆-E receptors (K₃₆₈-E +Neto1: 6.99 ± 0.47 ***P=0.0002; K_{375/379/382}H₃₇₆-E +Neto1: 1.69 ± 0.15 *P=0.0326; K_{368/375/379/382}H₃₇₆-E +Neto1: 1.15 ± 0.20 P=0.6495) (**Figure 5C** [↗](#); **Figure 5-figure supplement 2B**; **Table 1** [↗](#)). Similarly, Neto2 also considerably increased the I_K/I_G values for mutant K₃₆₈-E, rescuing the kainate efficacy (K₃₆₈-E +Neto2: 2.13 ± 0.13 ***P=0.0007), while it does not seem to have any significant effect on the K_{375/379/382}H₃₇₆-E mutant (K_{375/379/382}H₃₇₆-E +Neto2: 0.86 ± 0.08 P=0.2045) (**Figure 5C** [↗](#); **Figure 5-figure supplement 2B**; **Table 1** [↗](#)).

Consistent with the observation above, our examination of the rectification index revealed a significant increase in outward current for the K₃₆₈-E mutant (K₃₆₈-E +Neto1: 2.84 ± 0.16 **P=0.0021). However, K_{375/379/382}H₃₇₆-E did not show any difference compared to wild-type

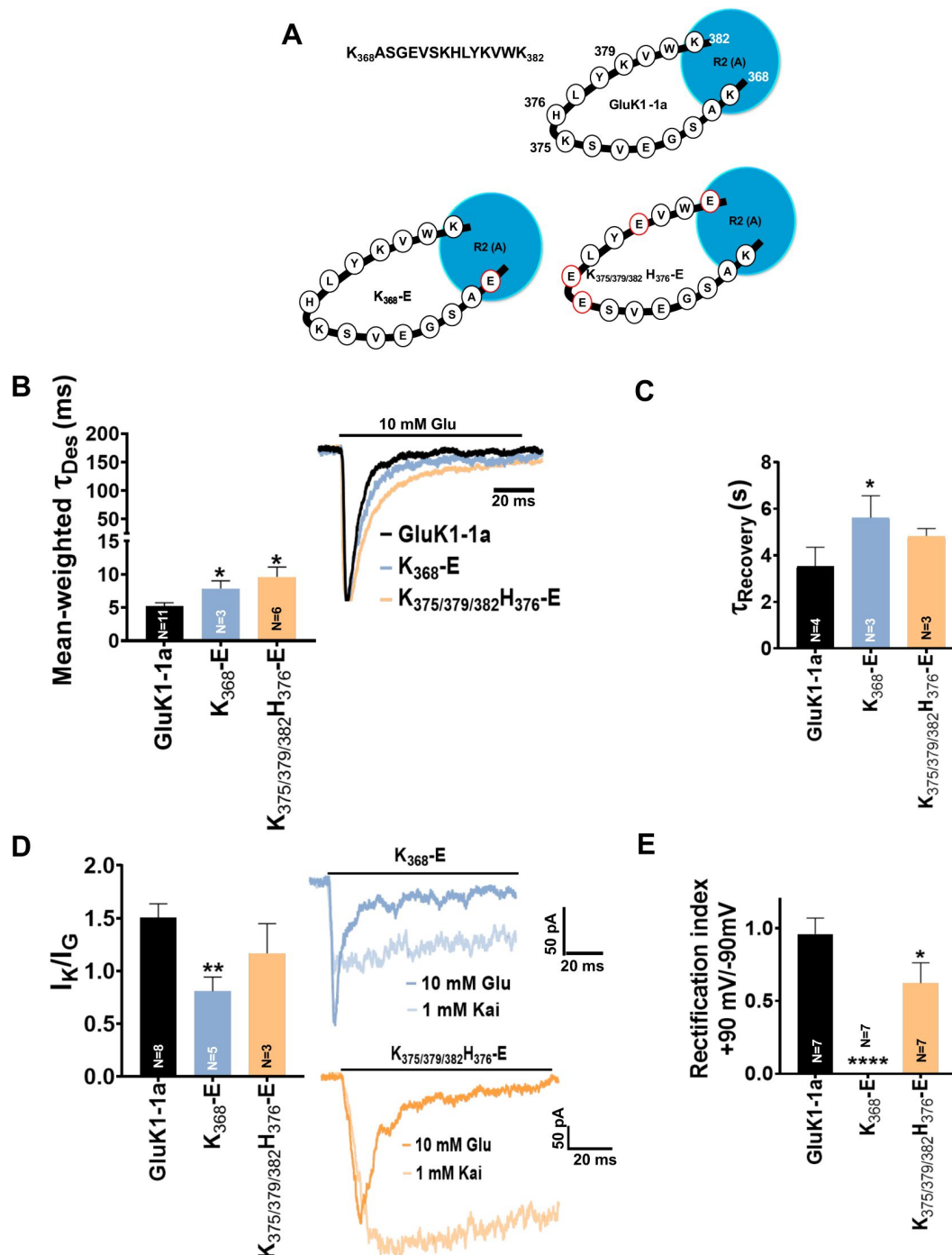


Figure 4.

Mutation of GluK1-1a splice insert residues affects the desensitization and recovery kinetics of the receptor. Bar graphs (mean $\pm$ SEM) show a comparison between wild-type and mutant receptors with or without Neto1 protein for different kinetic properties. **(A)** Schematic representation of 15 residues ATD splice ($K_{368}ASGEVSKHLYKVWK_{382}$) in wild-type and mutant receptors under study **(B)** Mean-weighted Tau (τ_{Des}) values for GluK1-1a wild-type and mutant receptors in the presence of 10 mM glutamate. **(C)** Tau ($\tau_{Recovery}$) recovery values for GluK1-1a and mutants. **(D)** The ratio of the peak amplitudes evoked in the presence of 1 mM kainate and 10 mM glutamate is shown for GluK1-1a mutants. **(E)** The rectification index represented by the ratio of currents evoked by 10 mM glutamate application at +90 mV and -90 mV for the wild-type and mutant receptors is shown. The wild-type GluK1 splice variant data is the same as from [Figure 1](#) and is replotted here for comparison. Error bars indicate mean $\pm$ SEM, N in each bar represents the number of cells used for analysis, and * indicates the significance at a 95% confidence interval.

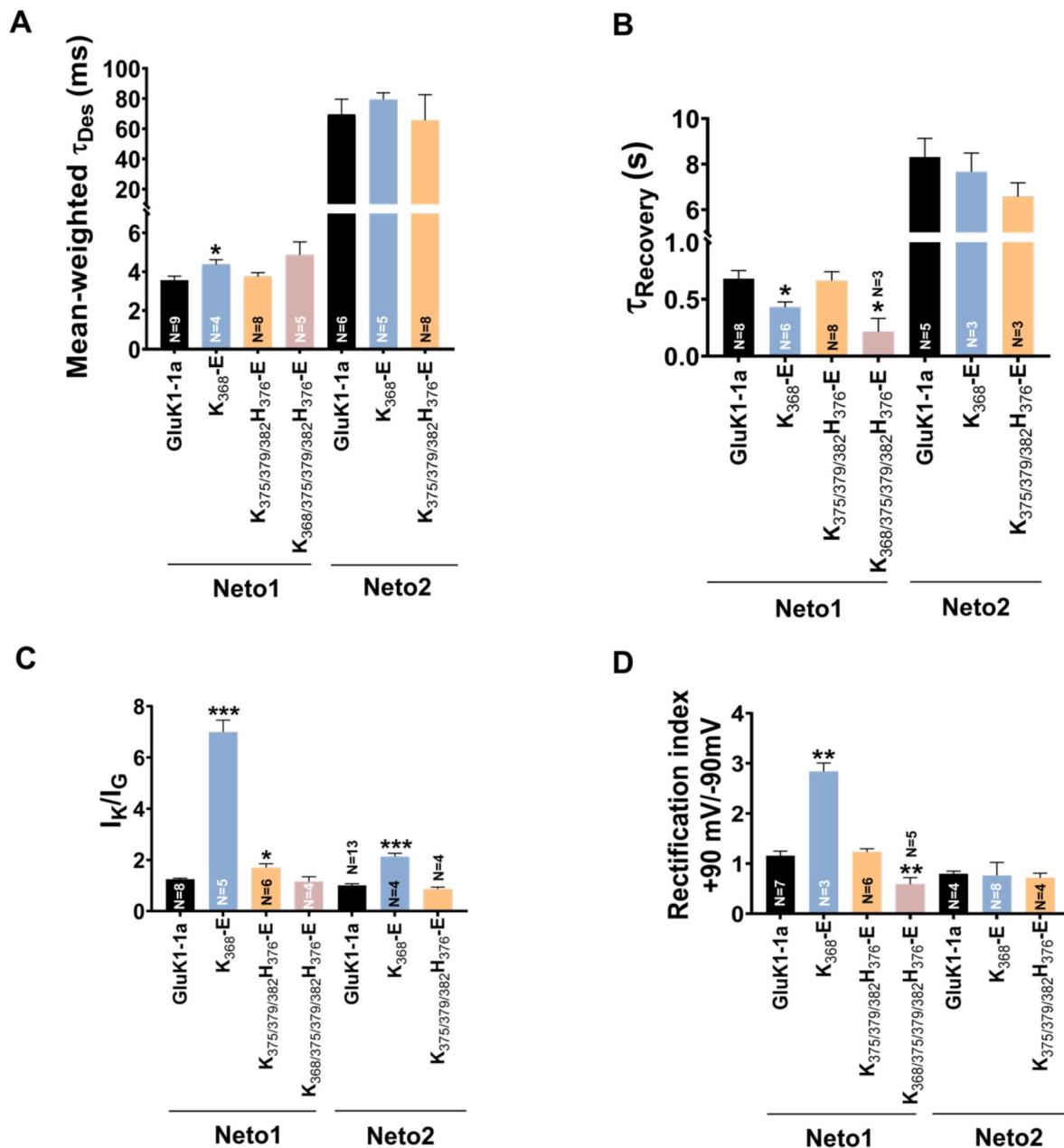


Figure 5.

Mutation of GluK1-1a splice insert residues affects the receptor modulation by Neto proteins. Bar graphs (mean $\pm$ SEM) show a comparison between wild-type and mutant receptors with Neto proteins for different kinetic properties. **(A)** Mean-weighted Tau (τ_{Des}) values for GluK1-1a wild-type and mutant receptors in the presence of 10 mM glutamate and expressed with Neto1/2. **(B)** Tau ($\tau_{Recovery}$) recovery values for GluK1-1a and mutants with Neto1/2. **(C)** The ratio of the peak amplitudes evoked in the presence of 1 mM kainate and 10 mM glutamate for GluK1-1a mutants co-expressed with Neto1/2 is shown. **(D)** The rectification index represented by the ratio of currents evoked by 10 mM glutamate application at +90 mV and -90 mV for the wild-type and mutant receptors with Neto proteins is shown. The wild-type GluK1 splice variants' data is the same as in **Figure 2** and is replotted here for comparison. Error bars indicate mean $\pm$ SEM, N in each bar represents the number of cells used for analysis, and * indicates the significance at a 95% confidence interval.

receptors, while $K_{368/375/379/382}H_{376}\text{-E}$ displayed a significant decrease in the outward current ($K_{375/379/382}H_{376}\text{-E} + \text{Neto1}$: 1.23 ± 0.07 $P=0.5351$; $K_{368/375/379/382}H_{376}\text{-E} + \text{Neto1}$: 0.59 ± 0.13 $**P=0.0081$) (**Figure 5D** [↗](#); **Figure 5-figure supplement 2C**; **Table 1** [↗](#)). Interestingly, however, the splice residue mutants did not show any significant variation in the rectification index when coexpressed with Neto2 (**Figure 5D** [↗](#); **Figure 5-figure supplement 2C**; **Table 1** [↗](#)).

Thus, our analysis of the gating properties of GluK1-1a mutants co-expressed with Neto proteins suggests that positively charged residues at positions 368, 375, 376, 379, and 382 in the splice insert influence receptor modulation by Neto proteins. Neto1 appears to have more pronounced effects on the mutant receptors compared to Neto2. Specifically, Neto1 significantly slowed desensitization for the $K_{368}\text{-E}$ mutant, accelerated recovery from desensitization for $K_{368}\text{-E}$ and $K_{368/375/379/382}H_{376}\text{-E}$ mutants, increased agonist efficacy for $K_{368}\text{-E}$ and $K_{375/379/382}H_{376}\text{-E}$ mutants, and altered rectification properties for $K_{368}\text{-E}$ and $K_{368/375/379/382}H_{376}\text{-E}$ mutants. In contrast, Neto2 had fewer significant effects on the mutant receptors, with the main impact being an increase in agonist efficacy for the $K_{368}\text{-E}$ mutant. Notably, Neto2 did not significantly affect desensitization, recovery from desensitization, or rectification properties of the mutant receptors when compared with wild-type GluK1-1a coexpressed with Neto2. These findings suggest that the splice residues in GluK1-1a differentially influence receptor modulation by Neto1 and Neto2, with Neto1 showing more extensive modulation of the mutant receptors' functional properties.

The structure of GluK1-1a_{EM} shows an overall conserved architecture of the desensitized state in kainate receptors

To evaluate the effects of splice residues on domain organization and structure of GluK1-1a receptors, we pursued its structure determination via single particle cryo-EM. Construct optimization was carried out to improve the expression and stability of purified protein. Briefly, the free cysteines in the TM1 region were mutated ($C_{552}Y$, $C_{557}V$) based on the sequence analysis with kainate and AMPA receptors, and this construct was named GluK1-1a_{EM} (**Figure 6-figure supplement 1 and 2**). The whole-cell patch clamp showed that GluK1-1a_{EM} was functional (**Figure 6-figure supplement 3**).

The structures of GluK1-1a_{EM} were determined using single-particle cryo-EM. The receptors were either DDM solubilized or reconstituted in lipid nanodiscs and trapped in a desensitized state using 2 mM of high-affinity agonist 2S, 4R-4-methyl glutamate (SYM2081) (**Figure 6-figure supplement 4**). A resolution of ~ 5.2 Å was achieved for the receptors in lipid nanodiscs, but the transmembrane region was not resolved due to an orientation bias (**Figure 6** [↗](#); **Figure 6-figure supplement 5**; **Figure 6-table supplement 1**). The full-length receptor in detergent micelles had a resolution of 8.2 Å, including the transmembrane region, which was ~ 8 Å for the extracellular domain (**Figure 6-figure supplement 5**; **Figure 6-table supplement 1**).

A tetrameric receptor model was built based on the crystal structures of GluK1-1a ATD, GluK1 LBD (kainate-bound state; PDB:3C32), and the TM domain based on a highly identical GluK2_{EM} (PDB:5KUF). Our cryo-EM map represented ATD residues-1-398, but the density corresponding to the ATD splice (368-382) was poorly resolved. The ATD-LBD linkers were resolved for all subunits (A to D) in both structures, the S1 and S2 domains were built entirely, and the TMD (TM1, TM3, and TM4) was built only for detergent-solubilized receptors (**Figure 6** [↗](#); **Figure 6-figure supplement 6 and 7**). For receptors reconstituted in lipid nanodiscs, we observed only the TM3 bundle. TM2 was not resolved in either dataset (**Figure 6** [↗](#); **Figure 6-figure supplement 7**). Extra densities were observed in the ECD layer of the GluK1-1a_{EM} ND map, which coincided with potential N-linked glycosylation sites (**Figure 6-figure supplement 8**). GluK1-1a_{EM} maps showed general conservation of the architecture of kainate receptors captured in the desensitized state (Khanra et al., 2021 [↗](#); Kumari et al., 2019 [↗](#); Meyerson et al., 2016 [↗](#); Selvakumar et al., 2021 [↗](#)). Consistent with earlier studies on homomeric and heteromeric KARs in the desensitized state, GluK1-1a exhibited a modular organization with three layers, namely, the two-fold symmetric ATD, quasi-

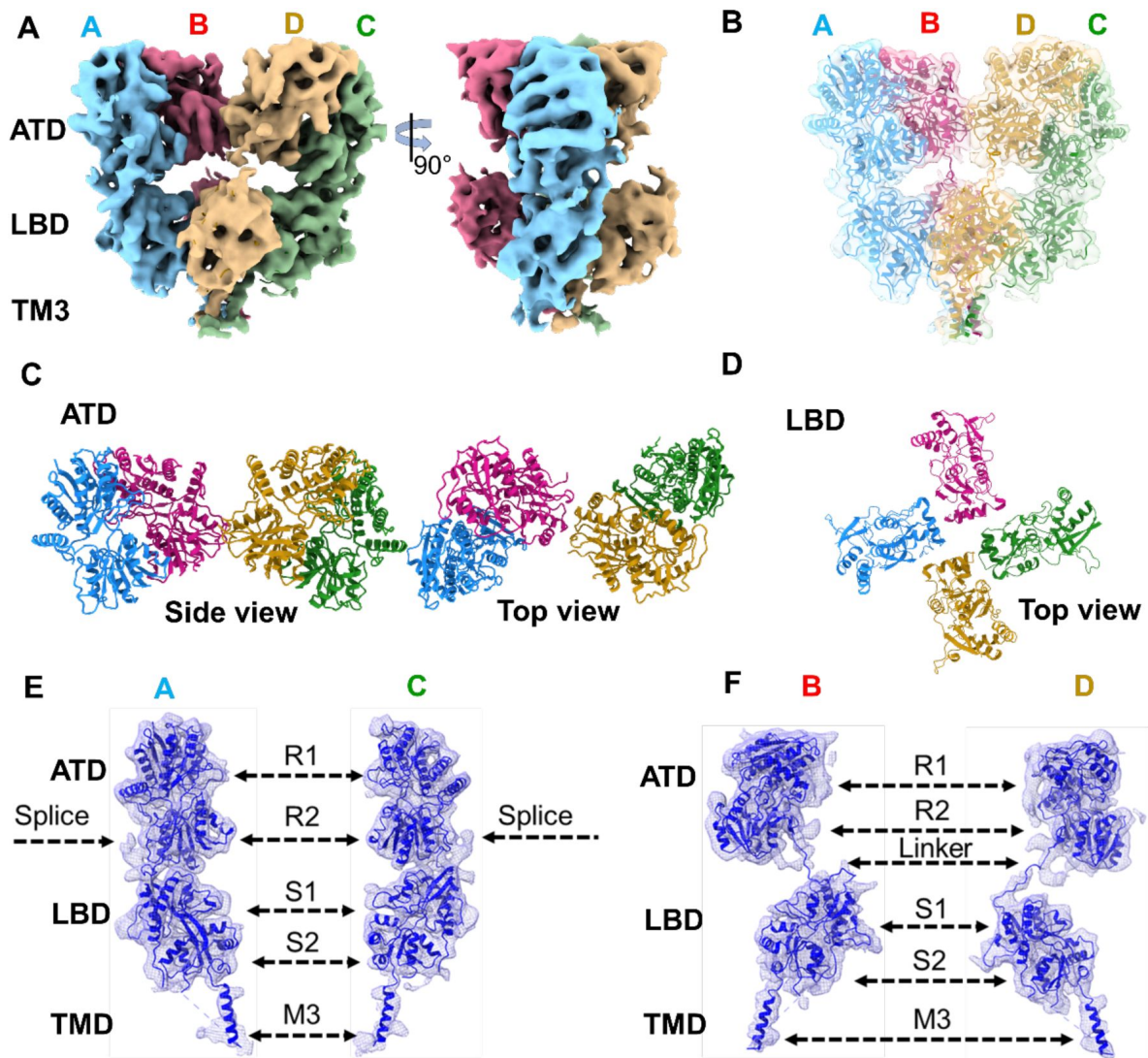


Figure 6.

Architecture of GluK1-1a_{EM} reconstituted in nanodisc for SYM-bound desensitized state. (A) Shows the segmented density map colored according to unique chains of the receptor tetramer (A-blue, B-pink, C-green, and D-gold) at 5.23 Å in side view and 90° rotated orientations. (B) Shows the final model fitted in the EM map. (C & D) Top views of ATD and LBD layers. (E & F) Display the segmented map fitted with the corresponding distal (A & C) and proximal (B & D) chains. Receptor sub-domains, the position of splice insertion, and linkers are indicated.

four-fold symmetric (LBD and four-fold symmetric TMD. The presence of the ATD splice insert in GluK1-1a did not affect the arrangement of the receptor domains in the desensitized state (**Figure 6**; **Figure 6-figure supplement 7**).

Discussion

Alternative splicing is a well-known mode of protein function modulation in various ion channels, such as transient receptor potential (TRP) (Gracheva et al., 2011; Zhou et al., 2013), big potassium (BK) (Chen et al., 2005), acid-sensing sodium (ASICs) (Bässler et al., 2001), and Shaker K⁺ channels (Hoshi et al., 1991). In the case of iGluRs, the 38 amino acid residue preceding TM4, known as flip and flop isoforms, is a well-characterized module of AMPARs that affects receptor expression and gating kinetics and is involved in various pathophysiology (Park et al., 2016; Sommer et al., 1990; Stine et al., 2001). For kainate receptors, the combination of different subunits (GluK1-GluK5) and post-transcriptional modifications, such as RNA editing and splicing, provides a broader range of pharmacological and gating properties, which can affect synaptic physiology (Kumar et al., 2011; Kumar and Mayer, 2010; Lerma, 2006; Straub et al., 2016). GluK1 is the most diversified subunit of KARs due to multiple post-transcriptional events and has been widely studied as a potential drug target. A recent report highlighted the equivalent presence of GRIK1-1 (exon 9) with respect to GRIK1-2 (lacking exon 9) (Herbrechter et al., 2021). The dominant presence of the GRIK1-1 gene was also reported in retinal Off bipolar cells of ground squirrel. Interestingly, an N-terminal splice (exon 5) in a similar position has been observed in the GluN1 subunit of NMDA receptors, which is known to influence the interactions at the ATD-LBD interface and, therefore, proton sensitivity and channel kinetics (Regan et al., 2018). However, this is the first report to decipher the role of the N-terminal splice insert in the kainate receptor family.

Our study compared the role of ATD splice in the kinetics of the GluK1-1a to a previously extensively characterized equivalent receptor without the splice region, GluK1-2a (Copits et al., 2011; Fisher, 2015; Fisher and Fisher, 2014; Sommer et al., 1992; Swanson and Heinemann, 1998). We found that the splice insert affects desensitization, recovery from desensitization, glutamate sensitivity, and channel rectification of the GluK1 receptor. Moreover, the splice also had a profound impact on desensitization with kainate, suggesting that the receptor follows different functional pathways for the two agonists. Our results showed a significant deviation from GluK1-2a in kainate-versus glutamate-evoked currents (I_K/I_G), implying a role for splice residues in imparting higher efficacy for kainate. This likely explains the enhanced stability of the kainate-bound state in GluK1-1a, leading to slower desensitization compared to GluK1-2a. Our functional assays of GluK1-1a receptors co-expressed with Neto proteins showed that Neto1 facilitates faster recovery from the desensitized state, consistent with previous reports (Copits et al., 2011; Fisher, 2015; Palacios-Filardo et al., 2016). However, we did not observe the fast onset of desensitization previously reported for GluK-2a (Copits et al., 2011; Fisher, 2015).

Additionally, our results show that Neto2 retards recovery from desensitization for both GluK1 variants, contradicting earlier reports of conflicting recovery rates with Neto2 for GluK1-2a. This indicates that Neto1 and Neto2 may interact with both GluK1 variants in a mutually exclusive manner or that the splice position may regulate the modulation behavior of Neto2 and Neto1. Our comparison of kainate and glutamate efficacies for the two variants also showed differential modulation by Neto proteins. Both Neto1 and Neto2 reduced the I_K/I_G ratios in GluK1-1a, while they significantly increased the I_K/I_G ratios in GluK1-2a, highlighting the impact of the splice insert on receptor response to the agonists (**Figure 3D**; **Figure 3-figure supplement 1B**; **Table 1**).

Our mutational analysis showed that K₃₆₈, a splice insert residue, affects channel kinetics. The charge reversal mutant (K₃₆₈-E) showed significant differences from the wild-type GluK1-1a for all tested properties, indicating a role for K₃₆₈ in protein-protein and/or protein-glycan interactions at

the ATD-LBD interface.

Our electrophysiological assays reveal a complex interplay between GluK1 receptor splice variants and Neto auxiliary proteins. While Neto2 demonstrates a more profound impact on wild-type GluK1-1a function compared to GluK1-2a, particularly in slowing desensitization, Neto1's modulation appears more sensitive to specific mutations in the splice insert. This suggests that Neto2 may have a stronger overall effect on receptor function through potentially allosteric mechanisms, whereas Neto1 might interact more directly with specific splice insert residues. The differential sensitivity to mutations highlights distinct roles for these auxiliary proteins: Neto2 as a powerful modulator of overall receptor function, and Neto1 as having more specific interactions with the splice insert region. This complexity underscores the nuanced nature of these protein interactions and the need for further structural and functional studies to fully elucidate the mechanisms underlying the functional diversity of GluK1 receptors.

Although our cryo-EM map poorly resolved the splice region, its position can be ascertained close to the ATD-LBD interface. Based on our functional assays, the splice seems to affect the interaction between the receptor and auxiliary proteins. The modulatory effects of Neto1 and Neto2 on GluK1 splice variants might be mediated by multiple conserved positively charged patches (Li et al., 2019; Vinnakota et al., 2021). The complex between GluK2-Neto2 (He et al., 2021b) provides a model that suggests K₁₈₃ and K₁₈₇ of GluK1 can potentially interact with a negative patch on Neto1 (D₁₄₀-E₁₄₄) and Neto2 (D₁₄₄-E₁₄₈). However, the splice insert appears to be positioned away from this interacting surface. Thus, it is possible that the splice residues may interact with Neto proteins when the receptor adopts a different conformation during the gating cycle. Additionally, splice residues could have an indirect or allosteric influence on the modulation of receptor functions and regulation by Neto proteins.

Our research, which encompasses structural, biochemical, biophysical, and functional investigations, highlights the significance of the unique N-terminal splice insert in the functions of GluK1 kainate receptors and opens avenues for further studies to understand its physiological effects. We also elucidated the important residues within the splice insert that could impact the modulatory behavior of auxiliary proteins. Our study emphasizes the need to investigate all possible combinations of KAR splice variants to better appreciate their contributions at different developmental stages. This comprehensive understanding of the distribution and functional diversity is essential for a rational therapeutic approach involving kainate receptors.

Materials and methods

Whole-cell patch-clamp electrophysiology

A whole-cell patch-clamp analysis was performed to understand the functional differences between GluK1-1a and GluK1-2a, as well as their modulation by Neto proteins. The constructs used for this study were HEK293 WT mammalian cells were seeded on siliconized glass coverslips in 35 mm dishes using Dulbecco's Modification of Eagle's Medium (DMEM) containing 10 % fetal bovine serum (FBS), 2 mM glutamine, and 10 units/mL Penicillin-Streptomycin. Cells were transfected with rat (*Rattus norvegicus*) *GRIK1-1a* or *GRIK1-2a* cloned in pRK7 (in the presence or absence of *rNeto1/rNeto2* cloned in IE-pRK8 as indicated) with EGFP expressing plasmid using Xfect (Clontech) according to manufacturer's instructions (receptor: Neto in 1: 3 and Receptor: EGFP in 1:1 DNA concentration). Similar protocols were followed to test the functionality of the *GRIK1-1a_{EM}* and *GRIK1-1a* splice mutant constructs. The whole-cell patch-clamp recording was performed with an EPC 10 USB amplifier (HEKA) 24-48 hours post-transfection. Cells were lifted using 1.5 mm diameter thin wall glass capillary tubes (30-0066 Harvard Apparatus), pulled to a fine tip with a Sutter P-1000 micropipette puller (Sutter Instruments, Novato, CA) containing internal solution (10 mM HEPES pH 7.4, 100 mM CsF, 30 mM CsCl, 4 mM NaCl, 0.5 mM CaCl₂, 5 mM

EGTA and 300 mOsm). Cells were continuously perfused with external/bath solution (10 mM HEPES pH 7.4, 150 mM NaCl, 2.8 mM KCl, 1.8 mM CaCl_2 , 1 mM MgCl_2 , and 300 mOsm) through a peristaltic perfusion system (Multichannel systems). Current was measured by holding the membrane at -60 mV, using a 2-kHz low-pass filter. Ligands (10 mM glutamate, 1 mM kainate, 2 mM SYM2081, or 10 μM UBP301) were applied through a double-barrelled theta pipette connected with an ultra-fast piezo-based perfusion system (Multichannel systems) via v8 Perfusion Fast-Step System (VC-77SP). Recordings were controlled and measured using Patchmaster-v2x90.2 (Heka Elektronik). The raw files were analyzed using Clampfit 11.2 (Molecular Devices) and Fitmaster-v2x90.4 (HEKA Elektronik). The data was fitted to logistic curves in GraphPad Prism (GraphPad Software Inc.). All traces were normalized before being used for calculations. The single exponential two-term fitting (Levenberg-Marquardt) was used to estimate the rate of desensitization (T_{Des}) for 100 ms glutamate application, measured as the decline of the current from 80% of its peak amplitude. Mean-weighted Tau (τ_{Des}) values were determined using the formula $[(\tau_1 \times \text{amplitude}_1) + (\tau_2 \times \text{amplitude}_2)] / [\text{amplitude}_1 + \text{amplitude}_2]$ where the tau values are the time constants from the exponential fit and the amplitudes are the assessed contributions of each component to the total peak current amplitude. To compute the recovery from desensitization, a series of paired-pulse experiments were performed with varying time pulses, where the amplitudes of the test pulse were normalized to that of the desensitizing pulse (calculated as relative amplitude) and plotted in comparison to the time (in seconds) between these two pulses. The recovery rate (T_{Recovery}) was obtained by fitting the one-phase association with an exponential function. Dose-response experiments were performed for GluK1-1a (co-expressed with or without Neto proteins) and GluK1-2a with different concentrations of glutamate or kainate in the range of 1 μM to 3 mM or 0.5 μM to 600 μM , respectively. Dose-response values were calculated as the percentage of maximum response against $\log[\text{agonist}]$ concentrations and fitted using variable slope (Hill's equation) in GraphPad Prism 8.0.1. For calculating kainate efficacy (I_K/I_G), the ratio of peak amplitudes obtained from the same cell evoked first by glutamate followed by kainate were employed. A ratio of peak amplitudes obtained at $+90/-90$ mV was utilized to calculate the rectification index. We investigated the voltage-dependent endogenous polyamine block by measuring current-voltage relationships. Current-voltage (IV) plots were prepared using the voltage ramp from -90 to $+90$ mV with a 10 mV increment step only after complete recovery of the receptor, and the current amplitude was normalized to that obtained at -90 mV. For kainate-evoked currents, the percentage desensitization was calculated. The steady-state current measured at the end of 1 s kainate application was divided by peak current. This ratio was then subtracted from 1 and multiplied by 100 to give percentage (%) desensitization (Fisher and Fisher, 2014 [\[3\]](#)).

Outside-out patch electrophysiology

HEK 293-T/17 cells were used for outside-out patch experiments 48-96 h post transfection. Cells were transfected with *GRIK1-1a* or *GRIK1-2a* cloned in pRK7 (in the presence or absence of rat (*Rattus norvegicus*) *rNeto1/rNeto2* cloned in IE-pRK8 as indicated) with EGFP expressing plasmid at DNA ratios of 3:0.5 for receptor alone and 2:8:0.5 with Netos respectively using TransIT®-LT1 Transfection Reagent according to manufacturer's instructions.

The extracellular solution used for the experiment contained 10 mM HEPES pH 7.3, 150 mM NaCl, 2.8 mM KCl, 2 mM CaCl_2 , 1 mM MgCl_2 , and 10 mM glucose. The intracellular solution contained 10 mM HEPES pH 7.3, 110 mM CsF, 30 mM CsCl, 5 mM EGTA, 4 mM NaCl, and 0.5 mM CaCl_2 .

Patch micropipettes were pulled using a P-97 Sutter puller, and the resistance was maintained at 2.3-2.6 megaohms. After forming the whole-cell configuration, the patch was pulled away from the cell to facilitate an outside-out patch. The cells were voltage-clamped at -70 mV, and a saturated concentration of glutamate (10 mM) was rapidly applied using a three-barrelled theta glass attached to a Siskiyou MXPZT-300 solution switcher. Glutamate applications of 1 ms were used to determine deactivation, while 100 or 1000 ms applications were used to assess desensitization.

Statistical analysis

Comparisons between wild-type receptors, EM, or mutant constructs were obtained using an unpaired *t*-test (two-tailed, with or without the Welch test) or Brown-Forsythe and Welch ANOVA, followed by Dunnett's multiple comparisons. Statistical analysis was carried out in GraphPad Prism, version 8.0.1. P values < 0.05 were considered statistically significant and are reported (* = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.001$, **** = $p < 0.0001$).

Site-Directed Mutagenesis (SDM)

Based on our electrophysiology analysis of the wild-type GluK1-1a/GluK1-2a receptors, structural analysis of GluK1-1a_{EM}, and recent reports that suggest that the presence of the positive patches in GluK2 ATD affects the interaction with Neto proteins (He et al., 2021b; Li et al., 2019; Vinnakota et al., 2021), we performed SDM to understand the role of splice residues in the receptor kinetics. All splice mutations were introduced in the wild-type (species) *GRIK1-1a* pRK7 construct for electrophysiology, as well as the *GRIK1-1a*_{EM}-EGFP-His₈-pEGBacMam construct for surface expression and pull-downs using the (overlap PCR) ligation-free cloning approach (Zhang et al., 2017). In brief, we performed the 2 sets of PCR using standard cloning primers (~300 bp upstream and downstream of the mutation) and the mutant primers as listed in Supplemental Table 1 to obtain the fragment containing our mutation of interest and flanking regions corresponding to the wild-type GluK1-1a. Next, the obtained fragment was used as a megaprimer to amplify the complete GluK1-1a vector backbone that would now contain the mutation. This PCR product was DpnI digested for 1h at 37 °C to remove parental DNA and was transformed into DH5 α strain of *E.coli*. Clones were confirmed by sequencing. Initially, the positively charged and other residues of the splice were substituted with alanine. Later, charge reversal mutants (K and/or H to E) were also prepared to understand their role in the receptor kinetics as well as interaction with Neto proteins. The mutants prepared are summarized in **Figure 4-table supplement 1**.

Co-immunoprecipitation (*in vitro*)

To understand the effect of mutations on GluK1-1a and Neto1/2 interaction, co-immunoprecipitation was carried out using an anti-His monoclonal antibody against the receptor. Similar constructs and transfection protocols were followed for surface expression analysis. Cells were pelleted 65-70 h post-transfection and washed with TBS (20 mM Tris pH 8, 150 mM NaCl). These cells were sonicated and solubilized in 500 μ L lysis buffer (20 mM Tris pH 8, 150 mM NaCl, 1% glycerol, protease inhibitor cocktail, 30 mM DDM). Post solubilization, debris was removed by centrifuging at 17,000 $\times$ g for 45 minutes at 4 °C. The supernatant was incubated for pre-clearing with 20 μ L of pre-equilibrated Protein A agarose beads (Thermo Scientific) for 1 h on a rotator at 4 °C. Post-pre-clearing, ~ 10% sample was saved as input, and the rest was used for pull-downs. Simultaneously, 2 μ g of an antibody (host: rabbit, Cat. No.- 12698, Cell Signaling Technology) was added to 40 μ L pre-equilibrated Protein A agarose (Thermo Scientific) and incubated for 1 h at 4 °C, followed by the addition of pre-cleared lysate. It was further incubated at 4 °C overnight (14-16 h). The unbound fraction was removed and washed four times with 500 μ L wash buffer (20 mM Tris pH 8, 150 mM NaCl, 1% glycerol, 0.75 mM DDM) to remove non-specific interactions. Protein was eluted in 30 μ L elution buffer (100 mM Tris pH 6.8, 12% glycerol, 4% SDS, 10 mM DTT, 2% β -mercaptoethanol) by heating at 95 °C for 10 minutes. Rabbit IgG controls were set up to confirm the validity of the experiment. To analyze the pull-down, 8% SDS-PAGE was used, followed by a western transfer. The immunoblots were probed using anti-actin (mouse, A3853, Sigma), anti-His (rabbit, 12698, Cell Signaling Technology), and anti-Neto1 (rabbit, SAB3500679, Sigma)/anti-GFP (mouse, G1546, Sigma) to detect the internal control (actin), the receptors and the co-immunoprecipitated Neto proteins respectively.

Construct design for expression and purification of rat GluK1-1a_{EM}

To obtain functional GluK1 receptors, rat (*Rattus norvegicus*) *GRIK1* with ATD splice insert and the shortest C-terminal domain, *GRIK1-1a* (1-871 amino acid residues), was cloned in the pEGBacMam vector. The receptor was cloned in-frame with a thrombin recognition site (LVPRGSAAAA), EGFP (A₂₀₇K; non-dimerizing mutant), and His₈ at the C-terminus. The wild-type protein generated very low amounts of the tetramer, as observed in fluorescence-assisted size-exclusion chromatography (FSEC). Therefore, based on the alignments of the sequence with GluK2_{EM}, GluK3_{EM} and GluA2, we mutated free cysteines in the TM1 region to residues corresponding to those of GluK2_{EM} or GluA2 [1x Cys (C₅₇₆S), 2x Cys (C₅₅₂Y, C₅₅₇V) and 3x Cys (C₅₅₂Y, C₅₅₇V, C₅₇₆S)] to obtain good yields of the tetrameric receptor. All clones were confirmed by restriction digestion and sequencing. The expression of all the mutants was confirmed by immunoblotting against His-tag and FSEC. GluK1-1a with 2x Cys mutations (C₅₅₂Y, C₅₅₇V) gave us the best receptor quality as observed in FSEC and, therefore, was used as GluK1-1a_{EM} for large-scale purification for structural studies.

GluK1-1a_{EM} expression and purification

Three liters of Human embryonic kidney (HEK) 293 GnTI⁻ suspension-adapted cultures (~1.5-2.0 x 10⁶ cells/mL) at 0.5 µg/mL were transfected (or infected with virus) with rat *GRIK1-1a_{EM}* plasmid [or baculovirus, prepared in DH10Bac as per established protocol (Goehring et al., 2014)] using polyethyleneimine (PEI-MAX, Polysciences; 1 DNA: 3 PEI w/w) as the transfection agent (or, infected at a multiplicity of infection of ~2). To boost the protein production, 10 mM sodium butyrate (Sigma) was added 16 h post-transfection/infection, and cultures were incubated at 30 °C for protein expression. Cells were harvested 65-70 h after transfection/infection, washed with buffer containing 20 mM Tris pH 8 and 150 mM NaCl, and stored at -80 °C until further processing. The cell pellet was resuspended (20 mL/L culture volume) in lysis buffer (20 mM Tris pH 8, 150 mM NaCl, and protease inhibitor cocktail, Roche). These resuspended cells were disrupted using sonication (QSonica sonicator, 3 cycles of 90 s; 10 s ON/ 20 s OFF; temperature cut-off: 15 °C). The lysate was clarified using low-speed spin (4307 x g for 20 minutes at 4 °C). The membranes were pelleted using ultracentrifugation (118991 x g for 1 h at 4 °C). Solubilization of membrane fraction was done in buffer containing 20 mM Tris pH 8, 150 mM NaCl, 30 mM DDM (Anatrace, D310LA), 5% glycerol, 10 mM imidazole, and protease inhibitor cocktail for 1 h on rotator at 4 °C. The non-solubilized fraction was separated by centrifugation at 47850 x g for 1 h at 4 °C and 4 mL of cobalt-charged TALON resin (Clontech, Takara) was added to the solubilized fraction for batch binding (3 h at 4 °C). Beads were harvested. The column was packed and washed with wash buffer (20 mM Tris pH 8, 150 mM NaCl, 1 mM DDM, 1% glycerol, 40 mM imidazole) until OD₂₈₀ reached zero. The bound receptor was eluted with elution buffer (20 mM Tris pH 8, 150 mM NaCl, 1 mM DDM, 1% glycerol, 250 mM imidazole). The peak fractions were pooled, concentrated at 1.3 mg / mL, and digested overnight at 4 °C with thrombin (1:100 w/w). Simultaneously, purified GluK1-1a_{EM} was also reconstituted in MSP1E3D1 nanodiscs with soybean polar lipids (Avanti Polar Lipids, 541602P) in a 1:2:140 ratio (GluK1-1a: MSP: lipids) following already established protocols (Chen et al., 2017; Gao et al., 2016; Ritchie et al., 2009). In brief, the purified GluK1-1a_{EM} in DDM was incubated with lipids, MSP1E3D1, 14 mM sodium cholate, and 1 mM PMSF for 30 min at 4 °C. The reconstitution of protein in nanodisc was initiated by removing detergent using equilibrated Bio-Beads SM-2 (Biorad, 150 mg/mL) for 4-6 h at 4 °C on an end-to-end rotator. The thrombin-digested GluK1-1a_{EM} receptors in DDM, or nanodisc-reconstituted receptors, were further purified via gel filtration (Superose 6 10/300, GE) in buffer containing 20 mM Tris pH 8, 150 mM NaCl, 0.5% glycerol, and 0.75 mM DDM (no detergent for nanodisc protein). The peak fractions were pooled and concentrated to ~0.6 mg/mL (final DDM concentration, ~7.5 mM) or ~0.9 mg/mL GluK1-1a_{EM} in nanodisc. All the affinity elution fractions and final purified protein were confirmed for purity and homogeneity using SDS-PAGE and FSEC, respectively. In the subsequent sections, purified GluK1-1a_{EM} in detergent and nanodisc will be called GluK1-1a_{EM} DDM and GluK1-1a_{EM} ND, respectively.

Cryo-EM sample preparation and data collection

Before grid preparation, the purified protein was incubated with 2 mM 2S, 4R-4-methyl glutamate (SYM2081) to capture the receptor in the desensitized state. The concentration of SYM2081 was confirmed by electrophysiology, and the stability of the receptor-SYM2081 complex was tested using FSEC.

GluK1-1a_{EM} DDM: Double application of 3 μ L protein ($\sim$ 0.6 mg/mL) SYM2081 complex was carried out on glow-discharged gold grids (R 0.6/1, 300 mesh, Quantifoil). The grids were blotted for 4.5 s and 3.5 s, respectively (0 blot force). Vitrobot temperature was maintained at 12 °C with 100% humidity, and the sample was vitrified in liquid ethane. The clipped grids were loaded into a 300 keV Titan Krios microscope equipped with a Falcon III direct detector camera (4k x 4k). Movies were recorded in counting mode with a nominal magnification of 59,000X (pixel size: 1.38 Å), and the defocus range of -2.0 to -3.2 μ m increased in steps of 0.3. Each movie comprises 25 frames with a total exposure of 60 seconds. A dose rate of 0.78 e⁻/frame was applied, with the total dose being 19.5 e⁻/Å².

GluK1-1a_{EM} ND: Double application of 2 μ L protein ($\sim$ 0.9 mg/mL) SYM2081 complex was carried out on glow-discharged gold grids (1.2/1.3, 200 mesh, Quantifoil). Grids were blotted for 3 s and 10 s, respectively, followed by vitrification. The clipped grids were loaded into a 300 keV Titan Krios microscope equipped with a K2 direct-detector camera (Gatan). The movies were recorded in super-resolution mode with an energy filter (20eV slit) and a pixel size of 1.41 Å. Each movie was composed of 30 frames with a total exposure of 12 seconds. A dose rate of 1.36 e⁻/frame was applied, with a total dose of 40.8 e⁻/Å².

Single-Particle Analysis

All movies (GluK1-1a_{EM} DDM-SYM: 1100, GluK1-1a_{EM} ND-SYM: 1535) were motion-corrected using UCSF Motioncor2 (Zheng et al., 2017 [\[1\]](#)). Bad micrographs were removed manually post contrast transfer function (CTF) estimation using CTFFIND4 (Rohou and Grigorieff, 2015 [\[2\]](#)). Manually picked particles ($\sim$ 1000) were 2D classified and used as templates for autopicking in RELION or cryoSPARCv3. For GluK1-1a_{EM} DDM-SYM and GluK1-1a_{EM} ND-SYM, the autopicked particles in cryoSPARCv3 were cleaned up with multiple rounds of 2D classification. Particles in the best classes were used for training TOPAZ for automated particle picking. For GluK1-1a_{EM} DDM-SYM, initially, 13,750 particles were picked; post iterative rounds of 2D and 3D classification, 5372 particles were used for final 3D reconstruction. In the case of GluK1-1a_{EM} ND-SYM, initially, 1,97,908 particles were picked. Multiple rounds of clean-up using 2D and 3D classification gave 24,531 particles that were used for the final 3D reconstruction.

The final particles were corrected for local motion in cryoSPARCv3, followed by refinement using C1 symmetry for both GluK1-1a_{EM} DDM-SYM and GluK1-1a_{EM} ND-SYM. Since we had less information on TMD in both the detergent and nanodisc forms, as observed in 2D classes, we performed local refinement for ATD and LBD using an ECD mask for GluK1-1a_{EM} DDM-SYM and GluK1-1a_{EM} ND-SYM, which improved map resolution to 8.01 Å and 5.2 Å (0.143 FSC) respectively. For GluK1-1a_{EM} DDM-SYM, local refinement using a mask for full-length structure yielded a resolution of 8.2 Å. The 3D maps were sharpened via cryoSPARCv3 (Punjani et al., 2017 [\[3\]](#)) or DeepEMhancer (Sanchez-Garcia et al., 2021 [\[4\]](#)) using a mask from the cryoSPARCv3 refinement output. Local Resolution was estimated using BlocRes (Cardone et al., 2013 [\[5\]](#)) in cryoSPARCv3 or Phenix1.19.2 (Afonine et al., 2018a [\[6\]](#)).

Model building

Tetrameric assembly for the GluK1-1a_{EM} ND-SYM model was built using crystal structures of individual domains (ATD, not yet published, and LBD, PDB: 3C32, GluK1 LBD crystal structure with kainate) fitted into the EM map (5.2 Å) in UCSF Chimera (Pettersen et al., 2004 [↗](#)). Further, for building GluK1-1a_{EM} ND transmembrane domain 3 and GluK1-1a_{EM} DDM trans-membrane domain, GluK2 TMD (PDB:5KUF) was used due to high identity (~95%) for fitting into EM density. Phenix1.19.2 (Afonine et al., 2018b [↗](#)) and Namdinator (Kidmose et al., 2019 [↗](#)) were used to improve the model via the rigid body and Molecular Dynamics based flexible fitting, respectively. The final models fit well into the EM map of GluK1-1a_{EM} ND-SYM, GluK1-1a_{EM} DDM-SYM ECD, and FL, respectively, with a definite density for ATD, LBD, and TMD (pre-M1, M1, M3, and M4). Coot 0.9.4 (Emsley and Cowtan, 2004 [↗](#)) was used to analyze the final models, and chimera/chimeraX (Pettersen et al., 2021 [↗](#), 2004 [↗](#)) was used for figure preparation.

Supplementary Methods

Spatiotemporal distribution of exon 9 of GluK1 using transcriptomics data analysis

To understand the abundance of *GRIK1-1* splice in the human brain, we resorted to RNA-Seq data from the BrainSpan atlas that constitutes various databases to study transcriptional mechanisms in human brain development. The RNA-Seq data were downloaded for *GRIK1* (<https://www.brainspan.org/rnaseq/gene/1099967> [↗](#)), plotted in Excel (x-axis: regions of the brain; y-axes: expression in log₂ of Reads Per Kilobase per Million mapped reads, RPKM and age of donor). The heat maps were generated using the Excel file (CSV format) and RStudio (<https://www.rstudio.com/> [↗](#)) to determine the presence of GluK1 in various regions of the brain at different developmental stages, as shown in **Figure 1** [↗](#). Further, we narrowed it down to the *GRIK1-1* splice (exon 9; start position 30968845, 45 nucleotides in length) and tried to understand how it overlaps with the entire *GRIK1* gene expression. These data explored the presence of the ATD splice irrespective of which C-terminal splice variant is present.

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Author contributions

SD optimized GluK1-1a construct, expressed and purified protein, carried out molecular biology and biochemical experiments, and processed EM data, RV and PMC carried out all the electrophysiology experiments. Janesh Kumar supervised the overall project design and execution. SD, PMC, and JK analyzed data and wrote the manuscript. All authors approved the final draft.

Conceptualization: JK; Methodology: SD, PMC, RV, JK; Investigation: SD, PMC, RV, JK; Visualization: SD, PMC, RV, JK; Funding acquisition: JK; Project administration: JK; Supervision: JK; Writing – original draft: SD, PMC, JK; Writing – review & editing: SD, PMC, JK

Competing interests

Authors declare that they have no competing interests.

Data and materials availability

The cryo-EM density reconstructions and final models were deposited in the Electron Microscopy Data Base (accession codes EMD-34076, EMD-34083, and EMD-34197) and the Protein Data Bank (accession codes 7YSJ, 7YSV, and 8GPR). All other relevant data supporting the key findings of this study are available in the article and its [Supplementary Information](#) files or from the corresponding author upon request.

References

- Afonine P.V., Klaholz B.P., Moriarty N.W., Poon B.K., Sobolev O.V., Terwilliger T.C., Adams P.D., Urzhumtsev A (2018) **New tools for the analysis and validation of cryo-EM maps and atomic models** *Acta Crystallogr. Sect. Struct. Biol* **74**:814–840 <https://doi.org/10.1107/S2059798318009324>
- Afonine P.V., Poon B.K., Read R.J., Sobolev O.V., Terwilliger T.C., Urzhumtsev A., Adams P.D (2018) **Real-space refinement in PHENIX for cryo-EM and crystallography** *Acta Crystallogr. Sect. Struct. Biol* **74**:531–544 <https://doi.org/10.1107/S2059798318006551>
- Barbon A., Barlati S (2011) **Glutamate receptor RNA editing in health and disease.** *Biochem Biokhimiia* **76**:882–889 <https://doi.org/10.1134/S0006297911080037>
- Bässler E.L., Ngo-Anh T.J., Geisler H.S., Ruppersberg J.P., Gründer S (2001) **Molecular and functional characterization of acid-sensing ion channel (ASIC) 1b** *J. Biol. Chem* **276**:33782–33787 <https://doi.org/10.1074/jbc.M104030200>
- Bernard A., Khrestchatisky M (1994) **Assessing the extent of RNA editing in the TMII regions of GluR5 and GluR6 kainate receptors during rat brain development** *J. Neurochem* **62**:2057–2060
- Bettler B., Boulter J., Hermans-Borgmeyer I., O'Shea-Greenfield A., Deneris E.S., Moll C., Borgmeyer U., Hollmann M., Heinemann S (1990) **Cloning of a novel glutamate receptor subunit, GluR5: expression in the nervous system during development** *Neuron* **5**:583–595
- Bowie D (2008) **Ionotropic Glutamate Receptors & CNS Disorders** *CNS Neurol. Disord. Drug Targets* **7**:129–143
- Bowie D., Mayer M.L (1995) **Inward rectification of both AMPA and kainate subtype glutamate receptors generated by polyamine-mediated ion channel block** *Neuron* **15**:453–462 [https://doi.org/10.1016/0896-6273\(95\)90049-7](https://doi.org/10.1016/0896-6273(95)90049-7)
- Cardone G., Heymann J.B., Steven A.C (2013) **One number does not fit all: mapping local variations in resolution in cryo-EM reconstructions** *J. Struct. Biol* **184** <https://doi.org/10.1016/j.jsb.2013.08.002>
- Castillo P.E., Malenka R.C., Nicoll R.A (1997) **Kainate receptors mediate a slow postsynaptic current in hippocampal CA3 neurons** *Nature* **388**:182–186 <https://doi.org/10.1038/40645>
- Chen L., Tian L., MacDonald S.H.-F., McClafferty H., Hammond M.S.L., Huibant J.-M., Ruth P., Knaus H.-G., Shipston M.J (2005) **Functionally diverse complement of large conductance calcium- and voltage-activated potassium channel (BK) alpha-subunits generated from a single site of splicing** *J. Biol. Chem* **280**:33599–33609 <https://doi.org/10.1074/jbc.M505383200>
- Chen Q., She J., Zeng W., Guo J., Xu H., Bai X.-C., Jiang Y (2017) **Structure of mammalian endolysosomal TRPML1 channel in nanodiscs** *Nature* **550**:415–418 <https://doi.org/10.1038/nature24035>

- Copits B.A., Robbins J.S., Frausto S., Swanson G.T (2011) **Synaptic targeting and functional modulation of GluK1 kainate receptors by the auxiliary neuropilin and tolloid-like (NETO) proteins** *J. Neurosci. Off. J. Soc. Neurosci* **31**:7334–7340 <https://doi.org/10.1523/JNEUROSCI.0100-11.2011>
- Emsley P., Cowtan K (2004) **Coot: model-building tools for molecular graphics** *Acta Crystallogr. D Biol. Crystallogr* **60**:2126–2132 <https://doi.org/10.1107/S0907444904019158>
- Erreger K., Chen P.E., Wyllie D.J.A., Traynelis S.F (2004) **Glutamate receptor gating** *Crit. Rev. Neurobiol* **16**:187–224
- Fisher J.L (2015) **The auxiliary subunits Neto1 and Neto2 have distinct, subunit-dependent effects at recombinant GluK1- and GluK2-containing kainate receptors** *Neuropharmacology* **99**:471–480 <https://doi.org/10.1016/j.neuropharm.2015.08.018>
- Fisher M.T., Fisher J.L (2014) **Contributions of different kainate receptor subunits to the properties of recombinant homomeric and heteromeric receptors** *Neuroscience* **278**:70–80 <https://doi.org/10.1016/j.neuroscience.2014.08.009>
- Gao Y., Cao E., Julius D., Cheng Y (2016) **TRPV1 structures in nanodiscs reveal mechanisms of ligand and lipid action** *Nature* **534**:347–351 <https://doi.org/10.1038/nature17964>
- Goehring A., Lee C.-H., Wang K.H., Michel J.C., Claxton D.P., Bacongus I., Althoff T., Fischer S., Garcia K.C., Gouaux E (2014) **Screening and large-scale expression of membrane proteins in mammalian cells for structural studies** *Nat. Protoc* **9**:2574–2585 <https://doi.org/10.1038/nprot.2014.173>
- Gracheva E.O., Cordero-Morales J.F., González-Carcacia J.A., Ingolia N.T., Manno C., Aranguren C.I., Weissman J.S., Julius D (2011) **Ganglion-specific splicing of TRPV1 underlies infrared sensation in vampire bats** *Nature* **476**:88–91 <https://doi.org/10.1038/nature10245>
- Gregor P., O'Hara B.F., Yang X., Uhl G.R (1993) **Expression and novel subunit isoforms of glutamate receptor genes GluR5 and GluR6** *Neuroreport* **4**:1343–1346
- He L. *et al.* (2021) **Kainate receptor modulation by NETO2** *Nature* **599**:325–329 <https://doi.org/10.1038/s41586-021-03936-y>
- He L. *et al.* (2021) **Kainate receptor modulation by NETO2** *Nature* :1–5 <https://doi.org/10.1038/s41586-021-03936-y>
- Herbrechter R., Hübner N., Buchholz R., Reiner A (2021) **Splicing and editing of ionotropic glutamate receptors: a comprehensive analysis based on human RNA-Seq data** *Cell. Mol. Life Sci. CMLS* **78**:5605–5630 <https://doi.org/10.1007/s00018-021-03865-z>
- Hoshi T., Zagotta W.N., Aldrich R.W (1991) **Two types of inactivation in Shaker K⁺ channels: Effects of alterations in the carboxy-terminal region** *Neuron* **7**:547–556 [https://doi.org/10.1016/0896-6273\(91\)90367-9](https://doi.org/10.1016/0896-6273(91)90367-9)
- Huettnner J.E (2003) **Kainate receptors and synaptic transmission** *Prog. Neurobiol* **70**:387–407 [https://doi.org/10.1016/s0301-0082\(03\)00122-9](https://doi.org/10.1016/s0301-0082(03)00122-9)
- Jaskolski F., Coussen F., Nagarajan N., Normand E., Rosenmund C., Mülle C (2004) **Subunit composition and alternative splicing regulate membrane delivery of kainate receptors** *J. Neurosci. Off. J. Soc. Neurosci* **24**:2506–2515 <https://doi.org/10.1523/JNEUROSCI.5116-03.2004>

- Khanra N., Brown P.M., Perozzo A.M., Bowie D., Meyerson J.R (2021) **Architecture and structural dynamics of the heteromeric GluK2/K5 kainate receptor** *eLife* **10** <https://doi.org/10.7554/eLife.66097>
- Kidmose R.T., Juhl J., Nissen P., Boesen T., Karlsen J.L., Pedersen B.P (2019) **Namdinator - automatic molecular dynamics flexible fitting of structural models into cryo-EM and crystallography experimental maps** *IUCr* **6**:526–531 <https://doi.org/10.1107/S2052252519007619>
- Kumar J., Mayer M.L (2010) **Crystal structures of the glutamate receptor ion channel GluK3 and GluK5 amino-terminal domains** *J. Mol. Biol* **404**:680–696 <https://doi.org/10.1016/j.jmb.2010.10.006>
- Kumar J., Schuck P., Mayer M.L (2011) **Structure and assembly mechanism for heteromeric kainate receptors** *Neuron* **71**:319–331 <https://doi.org/10.1016/j.neuron.2011.05.038>
- Kumari J., Vinnakota R., Kumar J (2019) **Structural and Functional Insights into GluK3-kainate Receptor Desensitization and Recovery** *Sci. Rep* **9** <https://doi.org/10.1038/s41598-019-46770-z>
- Lerma J (2006) **Kainate receptor physiology** *Curr. Opin. Pharmacol* **6**:89–97 <https://doi.org/10.1016/j.coph.2005.08.004>
- Lerma J (2003) **Roles and rules of kainate receptors in synaptic transmission** *Nat. Rev. Neurosci* **4**:481–495 <https://doi.org/10.1038/nrn1118>
- Lerma J., Marques J.M (2013) **Kainate receptors in health and disease** *Neuron* **80**:292–311 <https://doi.org/10.1016/j.neuron.2013.09.045>
- Lerma J., Paternain A.V., Rodríguez-Moreno A., López-García J.C (2001) **Molecular physiology of kainate receptors** *Physiol. Rev* **81**:971–998
- Li Y.-J. *et al.* (2019) **Neto proteins regulate gating of the kainate-type glutamate receptor GluK2 through two binding sites** *J. Biol. Chem* **294**:17889–17902 <https://doi.org/10.1074/jbc.RA119.008631>
- Meyerson J.R., Chittori S., Merk A., Rao P., Han T.H., Serpe M., Mayer M.L., Subramaniam S (2016) **Structural basis of kainate subtype glutamate receptor desensitization** *Nature* **537**:567–571 <https://doi.org/10.1038/nature19352>
- Palacios-Filardo J., Aller M.I., Lerma J (2016) **Synaptic Targeting of Kainate Receptors** *Cereb. Cortex* **26**:1464–1472 <https://doi.org/10.1093/cercor/bhu244>
- Park Y.H., Broyles H.V., He S., McGrady N.R., Li L., Yorio T (2016) **Involvement of AMPA Receptor and Its Flip and Flop Isoforms in Retinal Ganglion Cell Death Following Oxygen/Glucose Deprivation** *Invest. Ophthalmol. Vis. Sci* **57**:508–526 <https://doi.org/10.1167/jiovs.15-18481>
- Perrais D., Coussen F., Mulle C (2009) **Atypical functional properties of GluK3-containing kainate receptors** *J Neurosci* **29**:15499–510
- Pettersen E.F., Goddard T.D., Huang C.C., Couch G.S., Greenblatt D.M., Meng E.C., Ferrin T.E (2004) **UCSF Chimera--a visualization system for exploratory research and analysis** *J. Comput. Chem* **25**:1605–1612 <https://doi.org/10.1002/jcc.20084>

- Pettersen E.F., Goddard T.D., Huang C.C., Meng E.C., Couch G.S., Croll T.I., Morris J.H., Ferrin T.E (2021) **UCSF ChimeraX: Structure visualization for researchers, educators, and developers** *Protein Sci. Publ. Protein Soc* **30**:70–82 <https://doi.org/10.1002/pro.3943>
- Pinheiro P., Mulle C (2006) **Kainate receptors** *Cell Tissue Res* **326**:457–482 <https://doi.org/10.1007/s00441-006-0265-6>
- Punjani A., Rubinstein J.L., Fleet D.J., Brubaker M.A (2017) **cryoSPARC: algorithms for rapid unsupervised cryo-EM structure determination** *Nat. Methods* **14**:290–296 <https://doi.org/10.1038/nmeth.4169>
- Regan M.C., Grant T., McDaniel M.J., Karakas E., Zhang J., Traynelis S.F., Grigorieff N., Furukawa H (2018) **Structural Mechanism of Functional Modulation by Gene Splicing in NMDA Receptors** *Neuron* **98**:521–529 <https://doi.org/10.1016/j.neuron.2018.03.034>
- Ren Z., Riley N.J., Needleman L.A., Sanders J.M., Swanson G.T., Marshall J (2003) **Cell surface expression of GluR5 kainate receptors is regulated by an Endoplasmic Reticulum retention signal** *J. Biol. Chem* <https://doi.org/10.1074/jbc.M309585200>
- Ritchie T.K., Grinkova Y.V., Bayburt T.H., Denisov I.G., Zolnerciks J.K., Atkins W.M., Sligar S.G (2009) **Chapter 11 - Reconstitution of membrane proteins in phospholipid bilayer nanodiscs** *Methods Enzymol* **464**:211–231 [https://doi.org/10.1016/S0076-6879\(09\)64011-8](https://doi.org/10.1016/S0076-6879(09)64011-8)
- Rohou A., Grigorieff N (2015) **CTFFIND4: Fast and accurate defocus estimation from electron micrographs** *J. Struct. Biol* **192**:216–221 <https://doi.org/10.1016/j.jsb.2015.08.008>
- Sanchez-Garcia R., Gomez-Blanco J., Cuervo A., Carazo J.M., Sorzano C.O.S., Vargas J (2021) **DeepEMhancer: a deep learning solution for cryo-EM volume post-processing** *Commun Biol* **4** <https://doi.org/10.1038/s42003-021-02399-1>
- Seeburg P.H (2002) **A-to-I editing: new and old sites, functions and speculations** *Neuron* **35**:17–20
- Selvakumar P., Lee J., Khanra N., He C., Munguba H., Kiese L., Broichhagen J., Reiner A., Levitz J., Meyerson J.R. (2021) **Structural and compositional diversity in the kainate receptor family** *Cell Rep* **37** <https://doi.org/10.1016/j.celrep.2021.109891>
- Sheng N., Shi Y.S., Lomash R.M., Roche K.W., Nicoll R.A (2015) **Neto auxiliary proteins control both the trafficking and biophysical properties of the kainate receptor GluK1** *eLife* **4** <https://doi.org/10.7554/eLife.11682>
- Sommer B., Burnashev N., Verdoorn T.A., Keinänen K., Sakmann B., Seeburg P.H (1992) **A glutamate receptor channel with high affinity for domoate and kainate** *EMBO J* **11**:1651–1656
- Sommer B., Keinänen K., Verdoorn T.A., Wisden W., Burnashev N., Herb A., Köhler M., Takagi T., Sakmann B., Seeburg P.H (1990) **Flip and flop: a cell-specific functional switch in glutamate-operated channels of the CNS** *Science* **249**:1580–1585 <https://doi.org/10.1126/science.1699275>
- Stine C.D., Lu W., Wolf M.E (2001) **Expression of AMPA receptor flip and flop mRNAs in the nucleus accumbens and prefrontal cortex after neonatal ventral hippocampal lesions** *Neuropsychopharmacol. Off. Publ. Am. Coll. Neuropsychopharmacol* **24**:253–266 [https://doi.org/10.1016/S0893-133X\(00\)00212-8](https://doi.org/10.1016/S0893-133X(00)00212-8)

Straub C. *et al.* (2016) **Distinct Subunit Domains Govern Synaptic Stability and Specificity of the Kainate Receptor** *Cell Rep* **16**:531–544 <https://doi.org/10.1016/j.celrep.2016.05.093>

Swanson G.T., Heinemann S.F. (1998) **Heterogeneity of homomeric GluR5 kainate receptor desensitization expressed in HEK293 cells** *J. Physiol* **513**:639–646 <https://doi.org/10.1111/j.1469-7793.1998.639ba.x>

Traynelis S.F., Wollmuth L.P., McBain C.J., Menniti F.S., Vance K.M., Ogden K.K., Hansen K.B., Yuan H., Myers S.J., Dingledine R. (2010) **Glutamate receptor ion channels: structure, regulation, and function** *Pharmacol. Rev* **62**:405–496 <https://doi.org/10.1124/pr.109.002451>

Valbuena S., Lerma J. (2021) **Losing balance: Kainate receptors and psychiatric disorders comorbidities** *Neuropharmacology* **191** <https://doi.org/10.1016/j.neuropharm.2021.108558>

Vinnakota R., Dhingra S., Kumari J., Ansari M.Y., Shukla E., Nerkar M.D., Kumar J. (2021) **Role of Neto1 extracellular domain in modulation of kainate receptors** *Int. J. Biol. Macromol. S* **141**:8130 <https://doi.org/10.1016/j.ijbiomac.2021.10.001>

Zhang H., Liu C.-J., Jiang H., Zhou L., Li W.-Y., Zhu L.-Y., Wu L., Meng E., Zhang D.-Y. (2017) **Mega primer-mediated molecular cloning strategy for chimaeragenesis and long DNA fragment insertion** *Biosci. Rep* **37** <https://doi.org/10.1042/BSR20160608>

Zhang W. *et al.* (2009) **A transmembrane accessory subunit that modulates kainate-type glutamate receptors** *Neuron* **61**:385–396 <https://doi.org/10.1016/j.neuron.2008.12.014>

Zheng S.Q., Palovcak E., Armache J.-P., Verba K.A., Cheng Y., Agard D.A. (2017) **MotionCor2: anisotropic correction of beam-induced motion for improved cryo-electron microscopy** *Nat. Methods* **14**:331–332 <https://doi.org/10.1038/nmeth.4193>

Zhou Y., Suzuki Y., Uchida K., Tominaga M. (2013) **Identification of a splice variant of mouse TRPA1 that regulates TRPA1 activity** *Nat. Commun* **4** <https://doi.org/10.1038/ncomms3399>

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Reviewer #1 (Public Review):

Kainate receptors play various important roles in synaptic transmission. The receptors can be divided into low affinity kainate receptors (GluK1-3) and high affinity kainate receptors (GluK4-5). The receptors can assemble as homomers (GluK1-3) or low-high affinity heteromers (GluK4-5). The functional diversity is further increased by RNA splicing. Previous studies have investigated C-terminal splice variants of GluK1, but GluK1 N-terminal (exon 9) insertions have not been previously characterized. In this study Dhingra *et al* investigate the functional implications of a GluK1 splice variant that inserts a 15 amino acid segment into the extracellular N-terminal region of the protein using whole-cell and excised outside-out

electrophysiology.

The authors convincingly show that the insertion profoundly impacts the function of GluK1-1a - the channels that have the insertion are slower to desensitize. The data also shows that the insertion changes the modulatory effects of Neto proteins, resulting in altered rates of desensitization and recovery from desensitization. To determine the mechanism by which the insertion exerts these functional effects, the authors perform pull-down assays of Neto proteins, and extensive mutagenesis on the insert.

The electrophysiological part of the study is very rigorous and meticulous.

The biggest weakness of the manuscript is the structural work. Due to issues with preferred orientation (a common problem in cryo-EM), the 3D reconstructions are at a low resolution (in the 5-8 Å range) and cannot offer much mechanistic insight into the effects of the insertion. Based on the available data, the authors posit that the insertion does not change the arrangement of the subunits in the desensitized state. However, there is no comparison with a structure that does not contain the insertion, so while the statement may well be true, no data is shown to support it.

Overall, the cryo-EM contributes little and distracts from the good parts of the manuscript.

Another part that does not contribute much is the RNAseq data that has been pulled from a database and analyzed for the paper. It is being used to show that the exon 9 insertion variant is predominantly expressed in the cerebellar cortex at early stages of brain development. The methods do not describe in detail how the data has been analyzed (e.g., is the data scaled per sample/gene or globally?) so it is hard to know what we can compare in the heat plots. In Figure 1- supplement 1 there aren't striking differences in expression (at least not obviously visible in the current illustration).

Despite these weaknesses, the study is a valuable contribution to the field because it characterizes a GluK1 variant that has not been studied before and highlights the functional diversity that exists within the kainate receptor family.

Revised manuscript:

The authors have clarified some of the issues raised by the reviewers, but no new data has been added to strengthen the manuscript. The structural part of the manuscript remains its weakest point, and the extent of mechanistic insight remains low.

<https://doi.org/10.7554/eLife.89755.2.sa3>

Reviewer #2 (Public Review):

Among ionotropic glutamate receptors, kainate receptors (KAR) are still the object of intense investigation to understand their role in normal and pathological excitatory synaptic transmission. Like other receptors, KAR appear under different splicing variants and their respective physiological function is still debated. In this manuscript, Dhingra et al explored the impact of the presence and of the absence of Exon9 of the GluK1 receptors on the pharmacological, biophysical and structural properties of the receptors. They further investigated how it is impacted by the association of KAR with their cognate auxiliary subunit Neto 1 and 2. This study represents a large body of work and data. The authors addressed the issue in a very systematic and rigorous manner.

First, by exploring RNAseq database, authors showed that GluK1 transcripts containing the exon 9 are present in many brain structures and especially in the cerebellum suggesting that a large part of GluK1 contains effectively this exon9.

Using HEK cells as an expression system, they characterized many gating and biophysical

properties of GluK1 receptors containing or not the exon9. Evaluated parameters were desensitization, relative potency of glutamate versus kainate, and polyamine block.

It is known that the association of GluK1 with auxiliary proteins Neto1/2 modulates the properties of the receptors. Authors investigated systematically whether Neto1 and 2 similarly alter GluK1 properties in function of the presence of exon9. This study provides many quantitative data that could be reused for modeling the role of kainate receptors. Given the change shown by the authors, the presence of exon in GluK1 is noticeable and likely should have an impact of synaptic transmission.

Interestingly, authors used a mutational approach to identify critical residues encoded by exon9 that are responsible for the functional differences between the two splice variants. In many cases, the replacement of a single amino acid leads to the absence of current confirming the crucial role of the segment of the receptor. However, it made the comparison and the identification of critical residues more challenging.

Authors attempted to establish the structure GluK1 receptors comprising the exon9 using different preparation methods. They succeeded in obtaining structures with equivalent or lower resolution compared with previous reports on GluK1 and GluK2 receptors. However, the organization of the peptide coded by exon is poorly defined and limited possible analyses. Despite this, they could observe that the presence of the exon9 does not alter significantly the structure of GluK1.

<https://doi.org/10.7554/eLife.89755.2.sa2>

Reviewer #3 (Public Review):

GluK1 forms glutamate-gated ion channels with an important function in synaptic transmission and neuronal excitability. Alternative RNA splicing has been described for these channels, allowing the diversification of GluK1 channels. The GluK1 splice variant GluK1-1a contains 15 residues in the amino-terminal domain, resulting from the Exon 9 splice insert. GluK1-1a displays significant expression in different regions of the brain, likely co-expressing with other GluK channels. The impact of the 15 residues on GluK1 channel properties and the overall structure has not been studied yet. The paper by Dhingra et al. aims to evaluate the impact of the Exon 9 splice insert on GluK1.1 channel function and structure. This work uses electrophysiological approaches, including whole-cell and patch clamp recordings, to determine the effect of the splice insert on GluK1.1 gating properties, including desensitization, agonist efficacy, recovery from desensitization, and rectification. By using mutagenesis and biochemical approaches, the authors studied the role of positively charged residues in the splice insert on channel properties and the interaction with modulatory Neto proteins. This work also shows the effect of the splice insert on the regulation of GluK1 channels by Neto proteins. Finally, by using Cryo-EM and single-particle analysis, the authors reconstructed a model for a homomeric GluK1-1a channel. Overall, this work provides two major milestones: 1) the first functional characterization of the GluK1-1a variant and 2) the first structure of this channel.

The functional data supporting the role of the insert on channel properties is convincing, although the current data does not provide significant insights about the mechanism. The overall structure in a putative desensitized state shows no differences with channels lacking the splice insert. However, some domains, including most of the 15 residues unique for the GluK1-1a variant were not resolved, suggesting high flexibility or conformational heterogeneity in those regions. Also, the low resolution of the obtained structures precludes conclusions on the structural basis for the role of the insert in channel function/regulation. Nonetheless, this paper represents an important advance in the study of glutamate receptors and invites the field to elucidate the structural basis for gating properties in GluK1-1a channels as well as other glutamate receptors. A more in-depth study about the role of

splicing variants on ligand binding affinity, regulation by modulatory subunits such as Neto proteins, and the potential impact of this specific variant on heteromeric channels would also be relevant.

<https://doi.org/10.7554/eLife.89755.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Recommendations For The Authors):

In its current form, I would exclude the cryo-EM data from the manuscript. It does not add much and it is distracting from the excellent work that you did on the functional characterization of the variant. Alternatively, you could try to improve the resolution and see if you can get some more meaningful analysis out of the structures? I noticed that you only collected very small datasets. If you decide to pursue a higher resolution reconstruction, collecting more movies will give you a better chance to obtain a higher resolution.

We express our gratitude to the reviewer for their invaluable feedback. While acknowledging that our structure currently maintains a low resolution, it still provides valuable insights into the splice's proximity to the N412 glycan density. This proximity and low-resolution map hindered the complete modeling of all the splice residues. Notably, this structure represents the first depiction of this particular splice variant. Consequently, it lays a foundation for subsequent studies in the field, and hence, we would want to keep it in the manuscript. As per reviewers' suggestions, we have now included comparisons of our structure with the GluK1-2a receptor structure reported recently (Mayerson et al. 2022). We do plan to carry out higher-resolution structures in the future.

I would probably also exclude the RNAseq analysis. I think that Figure 1 is fine, but the supplement 1 is not very successful in convincing me that the exon 9 is expressed mainly in early stages of brain development. In addition, the plot in Figure 1 indicates strong expression in the cerebellar cortex in 20s and 30s. If you decide to keep the data, I strongly encourage you to include more details on the analysis in the methods section.

Thanks for this insightful comment. We have now modified this section extensively for better clarity. Indeed, the expression of this variant seems to be dynamic in different brain regions. This has now been specified in the revised manuscript. Figure 1 shows the expression of GRIK1 exon 9 gene in different regions of the human brain and donor age. The supplementary figure 1 is a zoom-in on one such region, the Cerebral cortex, where we observe the maximum expression of GRIK1. In this region, we also observed higher expression of exon 9 in the early stages of development. The scales of Figure 1 (0-4 RPKM) and supplemental Figure 1 (0-6 RPKM) are different due to more expression of other exons in supplemental Figure 1 (example, we observe 4 RPKM expression in the shade of red, for figure 1, whereas similar values of 4 RPKM are orange-yellow in the supplemental figure 1). Using Supplemental Figure 1, we wanted to show the expression of exon 9 with respect to other exons during developmental stages that prove that GluK1-1 is highly expressed in the initial stages of life. more details on the analysis in the methods section has been added now.

Additionally, there are a few minor issues in the data presentation:

(1) in Fig. 2C there seems to be a mismatch between the green dose response plot and the GluK12a trace shown. The plot reports an EC50 of 187.7 uM, whereas in the sample

trace 0.25 mM agonist activates only to ~20%.

We have verified the data and statistics, confirming their consistency with the values reported in the manuscript. For Figure 2C, we present representative traces from a single cell. However, the EC50 value was calculated using Hill's equation based on averaged data from 5 cells.

(2) The axis label is misprinted in Figure 3C

Thanks. Corrected.

(3) In Fig 5 supplement 1, panel B - the 3 last labels above the western blot lanes are off so it is difficult to see which sample corresponds to which lane.

Thanks. We have corrected the figure.

Reviewer #2 (Recommendations For The Authors):

Overall I congratulate the authors of this study nicely done. It represents a large body of work.

We thank the reviewer for his/her time and positive comments.

I have several minor corrections that authors could consider for the revision of the manuscript P7. The desensitization rate of GluK1-2a was "delayed"... replace by "increased".

Corrected.

P9. Last line 0.37; P.. Add the P value.

P value has been added as suggested.

P11 authors indicate that K368/375//379/382H376-E mutant exhibit significant difference in desensitization properties in presence of Neto1, but on the 1st line of p11, they provide a P value above 0.05

We thank the reviewer for pointing out this discrepancy and have fixed the same. We have discussed two mutants that show slower desensitization when compared to GluK1-1a co-expressed with Neto1. The K to E mutant has significance, while the des value for the K368/375//379/382H376-E mutant shows the same pattern, though not significantly. We have now modified the text to explain this more clearly.

P19 the calculation of mean weighted tau TDes is not clear and should be better explained.

Thanks. We have added more details in the Methods sections. We analyzed the current decays in response to 1–2 ms or 1 s applications by employing an exponential function or the sum of two exponential functions. This analysis allowed us to derive a weighted mean τ_{des} using the formula $[(\tau_1 \times \text{amplitude}_1) + (\tau_2 \times \text{amplitude}_2)] / [\text{amplitude}_1 + \text{amplitude}_2]$. The tau values represent the time constants obtained from the exponential fits, while the amplitudes correspond to the estimated contributions of each component to the total peak current amplitude.

$$[(A1 * t1) + (A2 * t2)] / (A1 + A2)$$

It represents the calculation of a weighted mean, where A1 and A2 are the amplitudes, and t1 and t2 are the corresponding time constants. The formula calculates the overall mean time constant by taking into account the contribution of each component to the total amplitude.

P19 the rate of recovery was obtained by fitting the one-phase association "with" exponential function. With is missing.

We have corrected this error. Thanks.

P21 which method has been used for site directed mutagenesis

Overlapping PCR was carried out for mutagenesis using the primers listed in Figure 4-table supplement 1. A ligation-free cloning approach (Zhang et al., 2017) was used. It has now been elaborated in the methodology section under Site directed mutagenesis.

P21 and 22. Provide complete reference of reagent including species of antibodies.

Thanks. We have added all the details in the methods section now.

Anti-His: Rabbit mAb #12698 (Cell Signaling Technology)

Anti-Neto1: Rabbit #SAB3500679 (Sigma Aldrich)

Anti-GFP: Mouse mAb G1546 (Sigma Aldrich)

Anti-actin: Mouse mAb A3853 (Sigma Aldrich)

P22 How much anti His antibody was used with 40microliter of protein A?

We have used 2µg/ 40uL of Protein A slurry. This has now been added to the methodology.

P23 Authors seem to have used a virus to express protein but the protocol is not given. For example what is P2 virus?

We have now modified the manuscript to include details of baculovirus generation as per the protocol described in Goehring et al. 2014. We followed the same protocol wherein the 2nd generation of virus (P2) generated in insect (SF9) cells was used for infecting suspensionadapted HEK293-T cells for large-scale GluK1-1aEM protein expression.

Reviewer #3 (Recommendations For The Authors):

Major concerns:

(1) The effect of the splice insert on Gluk1 regulation by Neto proteins is not fully clear. For example, experiments in Fig. 3G indicate that the desensitization time for Gluk1-1a + Neto2 is ~32ms. This value is half compared with data obtained from whole-cell experiments shown in Fig. 3A (~70ms). What is the reason for this discrepancy? If variability is observed between experiments, I wonder how valid are the comparisons made in panel A between GluK11a+Neto2 vs GluK1-2a+Neto2 groups. In the case of recovery analysis, authors found significant differences comparing both groups in the presence of Neto (Fig. 3B) but recovery times are not identic for Gluk1-1a vs Gluk1-2a (without Neto). Thus, I wonder if the fold change related to the control group (without Neto) is different.

We appreciate your detailed feedback, which has allowed us to clarify and reinforce the validity of our experimental findings. Different recording configurations (e.g., outside-out patch (Fig. 3G) versus whole-cell recordings (Fig. 3A) have been used. Whole-cell recordings average responses over a larger membrane area and also have slower solution exchange times compared to outside-out patch recordings. This may have contributed to the variability in desensitization times. However, similar trends in our whole cell vs. outside-out patch recordings were observed. Further, all the data except those presented in Figs 3G and 3H are from whole-cell recordings. We have performed multiple independent experiments and utilized rigorous statistical analyses to validate our comparisons. We report mean values with standard deviations or confidence intervals to provide a more accurate representation of the data.

Neto1 significantly speeds up the recovery from desensitization for both variants, with a more pronounced effect on GluK1-1a (GluK1-1a +Neto1: 0.68 s) compared to GluK1-2a (GluK1-2a +Neto1: 1.15 s). The recovery times are not identical for the two variants, likely due to the presence of splice insert in GluK1-1a. Neto2, on the other hand, slows recovery for both variants without significant differential effects. However, the recovery rate from the desensitized state is faster for GluK1-1 compared to GluK1-2a alone, although insignificant (without Neto).

In the case of the glutamate concentration-response curve (Fig. 3C), EC50 values for Neto1 and Neto2 are relatively the same, but this approach on its own does not provide insights about the role of the splice insert. Previous experiments with the Gluk1 reveal differences between EC50 in the presence of Neto1 or 2 (Fisher, 2015), suggesting that the insert could regulate glutamate binding affinity, but still, this point is not directly demonstrated in this work.

Thanks for this insightful comment. Indeed, we cannot conclude that splice residues directly affect glutamate sensitivity and have modified the text accordingly. The Fisher paper demonstrated that both Neto1 and Neto2 can influence glutamate sensitivity in GluK1-2a, with EC50 values of $124.6 \pm 16.2 \mu\text{M}$. Specifically, in the presence of Neto1 and Neto2, the EC50 values are $4.4 \pm 0.4 \mu\text{M}$ and $13.7 \pm 4.2 \mu\text{M}$, respectively, indicating a noticeable effect though not substantially different for GluK1-2a coexpressed with either Neto1 and Neto2. Our observation for the GluK1-1a has been similar, with both Neto1 and Neto2 showing a leftward shift.

(2) Similar to the previous point, a proper interpretation of mutant data is missing in the manuscript. From current data, it is difficult to visualize the role of the insert on Netodependent regulation, mainly, because of the fact that some mutations alone affect Gluk1-1 channel properties. The authors conclude their data by stating that "while the modulation of the receptor by Neto 1 is affected by mutations in splice insert, the modulation by Neto 2 remains largely unaffected" (Page 13). However, this statement is confusing since the co-expression of Gluk1-1a with Neto2 (Fig. 5) prevents the effect caused by mutation K368 alone (Fig. 4), indicating that modulations by Neto 2 are indeed potentially affected by the mutations. Please, clarify. Also, the effect of the K368/375/379/382H376-E mutant on Neto modulation (pink bar in Fig. 5) is impossible to interpret properly since the effect of the mutation alone is not shown in the manuscript.

Thanks for seeking this important clarification. It is indeed true that splice residue mutations themselves affect the receptor functional properties in comparison to the wild-type receptors. For the sake of clarity, we have presented the effect of splice mutants on receptor properties separately from the effect of mutations on modulation by Neto proteins. Figure 4 demonstrates a comparison between wild-type and mutant receptors without the Neto

proteins, showcasing different kinetic properties, while Figure 5 provides detailed information on the role of the insert in Neto-dependent regulation.

It's true we could not record the effect of the K368/375/379/382H376-E mutant alone or when coexpressed with Neto 2 due to low peak amplitudes (mentioned in Table 1) that prevented reliable comparisons. However, robust currents were observed when the same mutant was coexpressed with Neto1, and hence comparisons were shown for this mutant with GluK1-1a wild-type + Neto1.

We have now modified the statement "while the modulation of the receptor by Neto 1 is affected by mutations in splice insert, the modulation by Neto 2 remains largely unaffected" and the last paragraph as follows:

"Neto1 appears to have more pronounced effects on the mutant receptors compared to Neto2. Specifically, Neto1 significantly slowed desensitization for the K368-E mutant, accelerated recovery from desensitization for K368-E and K368/375/379/382H376-E mutants, increased agonist efficacy for K368-E and K375/379/382H376-E mutants, and altered rectification properties for K368E and K368/375/379/382H376-E mutants. In contrast, Neto2 had fewer significant effects on the mutant receptors, with the main impact being an increase in agonist efficacy for the K368-E mutant. Notably, Neto2 did not significantly affect desensitization, recovery from desensitization, or rectification properties of the mutant receptors when compared with wildtype GluK1-1a coexpressed with Neto2. These findings suggest that the splice residues in GluK1-1a differentially influence receptor modulation by Neto1 and Neto2, with Neto1 showing more extensive modulation of the mutant receptors' functional properties."

(3) An open question after reading this interesting work is if the proposed change in Neto regulation because of the splice insert is due to changes in Gluk1-Neto interactions or because the rearrangement after interaction with Neto proteins is different. Pull-down experiments (Fig 5 Sup.1) suggest that the splice insert and all the mutants tested do not prevent interaction with Neto proteins. I wonder if the authors could complement their data with a quantitative approach/analysis to demonstrate if the splice insert and the mutants affect Neto1/2 interactions (as expected for the rationale when creating the mutants).

Thank you for this insightful suggestion. You raise an important point about distinguishing between changes in GluK1-Neto interactions and potential differences in receptor rearrangement after Neto binding. While our pull-down experiments suggest that the splice insert and mutants don't prevent Neto interactions (probably due to a larger interaction interface all along the receptor), a quantitative approach would indeed provide more nuanced information. In future studies, we do plan to perform a quantitative approach like Surface plasmon resonance to assess the changes in interactions upon mutations in the splice and/or Neto proteins in different states of the receptor. In addition, obtaining cryo-EM structures of GluK1 splice variants in complex with Neto1 and Neto2 would provide crucial insights into their interaction interfaces and any conformational changes induced by binding.

(4) Related to the Gluk1-1a structure, the authors state that the overall structure is similar to the one without the insert (page 14); however, this is not properly shown in the manuscript. Even if the overall architecture of the channel is the same, authors should make a proper/adequate comparison between both structures/domains to support their claims. Also, one should expect that the insertion of 15 amino acids would affect in some way the closing neighboring domains. The differential effect of the splice insert on glutamate and kainate EC50 values (Fig. 2 and Fig. 2 sup.1), suggests that the insert could introduce a sort of rearrangement in the binding domain. Thus, I wonder if a more elaborated analysis of the current structural data could reveal some structural insights

that would explain the specific functional differences due to the splice insert. If the low resolution and the missing residues avoid making some comparisons and establish differences between sidechain orientations, still, a proper comparison between the domain backbones would be helpful to validate the author's statement at least. Also, I wonder if the changes could be resolved better in a closed state or APO structure, instead of the desensitized structure. Finally, are the structures obtained in DDM and nanodiscs similar?

As per the reviewer's suggestion, we have now added a new figure in the supplementary information, "Figure 6-figure supplement 9," where we show a superimposition of GluK11aEM (detergent-solubilized or reconstituted in nanodiscs) and GluK1-2a (PDB:7LVT; silver) showing overall conservation of the structures in the desensitized state.

As evident from the figure and rmsd values mentioned above, we do not observe significant movements at both ATD and LBD layers of GluK1-1a with respect to GluK1-2a. Also as can be observed the DDM solubilized and nanodisc reconstituted GluK1-1a (Panel A) are very similar with a rmsd of ~2.19Å across all the 2664 Cα atom pairs. Due to low resolution of our structures, we have refrained from carrying out detailed structural comparisons.

Our efforts to capture the closed state or apo state structures have failed due to either severe orientation bias (only top views) or increased heterogeneity.

(5) Methods section lacks relevant information for proper data interpretation as well as for replicating some experiments in the future. For example:

A) The experimental design to determine the rectification index with a Ramp protocol is not clear: 1) Why the authors applied a ramp protocol if receptors desensitize along the time? Please clarify the protocol.

Ramp protocols were used only for the wild-type receptors to compare their voltage-dependent behavior, as this was the first study to compare the two splice variants. All kainate receptors (GluK1-GluK5) desensitize over time. However, their rectification properties have been studied previously (both the absence and presence of Neto proteins) using Ramp protocols as they are faster than step protocols.

B) Are polyamines included in the solutions to perform the rectification assays?

No, polyamines were not added to the intracellular solution, and the effect of the endogenous polyamine block was measured. This has now been specified in the results as well as the methods section.

C) It is not clear if the experiments to calculate IK/IG ratios were performed in the same preparation (This is, the same cell was stimulated with glutamate and then kainate or vice versa).

Indeed, the current responses for glutamate vs kainate are performed in the same cell (the same cell was stimulated by glutamate then kainate) so that the responses can be compared. It's now been specified in the methods section.

D) The experimental design for calculating recovery is not clear.

We employed a double pulse protocol to measure receptor recovery. The protocol involved applying two consecutive pulses of agonist stimulation to the receptor. Initially, we applied a brief agonist pulse to activate the receptor, followed by a specific recovery period. After the recovery period, we administered a second agonist pulse to assess the receptor's recovery

response. The receptor's recovery was determined by comparing the response amplitude of the second pulse to that of the first pulse, providing valuable insights into the receptor's recovery kinetics. Recovery rates were calculated with single exponential association fits in Prism. We have now modified the text for better clarity.

E) Please indicate the species used for both functional and Cryo-EM (rat Gluk1 isoform?).

Thanks for pointing this out. We have now specified in relevant methodology sections that *Rattus norvegicus* GluK1 and Neto proteins were used in this study.

F) Please describe the nanodisc reconstitution protocol and how the nanodisc protein was purified, if appropriate.

The MSP1E3D1 was purified by following the protocol given by the Sligar group in 2014 (doi: 10.1016/S0076-6879(09)64011-8). The nanodisc reconstitution protocol has now been elaborated in the revised manuscript.

G) Site-directed mutagenesis methodology is incomplete. Please check.

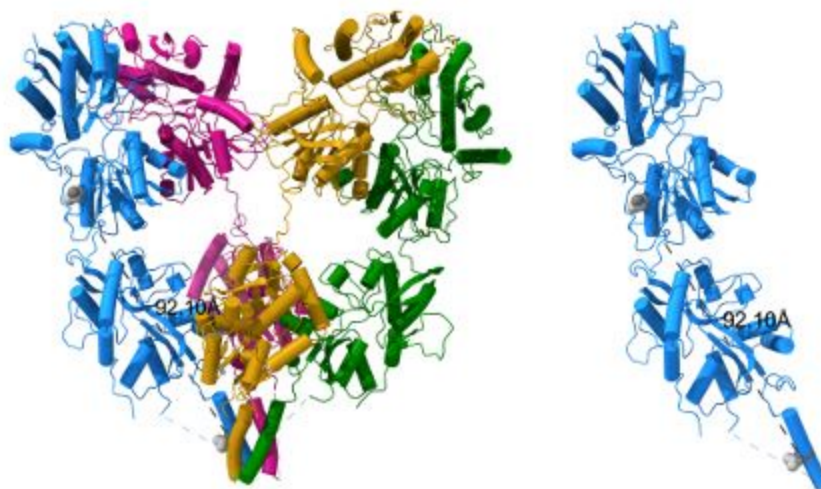
We have now elaborated this section to include more details.

Minor concerns:

(1) Authors state that splice residues are ~30Å away from the TM domain. Currently, there is no friendly representation showing the localization of the splice in the structure, besides Fig.6E. The manuscript could benefit itself if authors include a better 3D representation or a scheme to highlight the position of the splice relative to critical domains.

Thanks for pointing this out. The distance between TRP 381 CA (ATD) and LEU 636 CA (TM3) is 92.10 Å. We have changed the value in the text to ~92 Å.

Author response image 1.



(2) Authors mention that mutations in the insert to alanine show normal traffic to the plasma membrane but low current amplitude. Then, I wonder if single-channel conductance, mean open time or open probability is affected by the splice insert.

Showing the effects of the insert on single-channel properties would strengthen the manuscript's quality.

It is a good suggestion. However, as can be observed from our whole cell or outside out patch data, we obtained low peak amplitudes (<50 pA) for many of our receptor-only constructs and also suffered from high SEM for some recordings due to heterogeneity between cells of the same population. The suggestion to study the single channel properties of these receptors is considered for future experiments

(3) It is unclear how the insert or the mutations specifically affect glutamate- or kainate-induced responses because authors analyze IK/IG ratios only. Maybe authors could consider including an analysis of the role of the insert on specific glutamate- or kainate-induced response to gain insights about ligand selectivity.

All the values have been included in the excel for raw data. We have included the desensitization kinetics of mutant receptors in the presence of glutamate and compared it to the wild type GluK1-1a. Kainate induced responses were very heterogeneous (high SEM for % desensitization) and hence have not been included in the main data.

(4) Please be consistent with nomenclature along the manuscript to avoid confusion. For example, Are Gluk-1-1 and Gluk-1-1a referring to the same variant?

GluK1-1 has been used in the abstract and the introduction where we introduce the N-terminal splice variant which either has the 15 residues (termed as GluK1-1) or lacks it (GluK1-2). The C-terminal splice variants for GluK1 are named as "a-d", with "a" being the smallest C-terminal domain variant. Later in the manuscript, we have used only GluK1-1a terminology to represent the ATD splice variant with shortest C-terminal domain.

The introduction and spatiotemporal results talk about the GluK1-1 receptors wherein the

(5) Legend figure 2: Repeated phrase should be removed. Please check.

(6) Page 8: "This is similar to the effect observed in GluK1-2 receptors whereby the glutamate EC50 was shown to increase by Neto proteins [Neto1: 34-fold and Neto2: 7.5-fold (Palacios-Filardo et al., 2016) and Neto1/2: 10-30X (Fisher, 2015)]". It seems that values from Fisher's paper are backward. Please correct.

(7) Page 9. Second paragraph. Spelling mistake when referring to Fig. 3G.

Thanks for pointing out the inadvertent errors; we have now corrected all of them.

(8) Figure 3: The title in Y axis overlaps with the figure. Please check.

We have corrected the error.

(9) Page 10: "In addition, K375/379/382H376-E mutant also exhibited a slowdown in the recovery (K375/379/382H376-E: 4.83 {plus minus} 0.31 s P=0.2774) (Figure 4C; Table 1)." Statistical analysis indicates this is not correct. Please tone down this statement. For example: "...mutant also exhibited a trend to a slowdown in the recovery although differences do not reach statistical significance".

Thanks. We have modified the statement as suggested.

(10) Page 11: "and a reduction was observed for K375/379/382H376-E receptors (1.17 {plus minus} 0.28 P=0.3733) compared to wild-type (Figure 4D; Table 1)." Same issue as the previous minor comment.

Thanks. We have modified the statement as suggested.

(11) Page 11: "We observed that mutants K368-E and K368/375/379/382H376-E, desensitize significantly slower in the presence of Neto1" This statement is not true for K368/375/379/382H376-E mutant. Please correct.

Thanks. We have modified the statement as suggested and specified the difference.

(12) Legend Figure 4. Colored asterisks are not clear in the figure. Please check.

Thanks. The reference to colored asterisks has been removed from the legend as they are not used.

(13) Representative data shown in Fig 5 sup.2A do not match very well with the final quantification shown in Fig 5A. Please check. Also, the authors state in the result section (page 10) that data shown in Fig. 5A indicate that "GluK1-1a modulation by Neto 1 is influenced by the splice residues". This could be true only for residue K368; however, this is not so obvious since the two mutants containing K368E are inconsistent. Please check and clarify.

Only representative traces are shown in Fig 5 sup 2 A. However, the quantification shown in Fig 5 A is from multiple cells. We have rechecked all the data and found it to be consistent. We have rewritten this section and modified it for better clarity.

(14) Figure 6-supplement 2: Please incorporate missing values of MW standards in panel B.

Thanks. We have modified the figure to include values for MW standards.

(15) It is not clear the rationale for showing construct C552Y C557V C575S in Fig. 6 sup.3, panel A. This mutant is not mentioned in the manuscript.

It has been mentioned in the methodology section under "Construct design for expression and purification of rat GluK1-1aEM". It (C552Y C557V C576S) is one of the constructs used in optimizations that were checked for good protein yields. Based on FSEC protein profiles, we used C552Y, C557V (2X Cys mutant) as GluK1-1aEM, which is mentioned in the same section.

(16) Fig. 6 sup.4 Not clear what does mean w.r.c. Please specify in the legend.

With respect to (w. r. t.) has been specified in the manuscript.

(17) Suggestion to improve data presentation in Fig. 4D and Fig. 3 sup.1B: For easier comparison of IK/IG ratios, representative traces for kainate and glutamate in the same group could be shown using the same Y-scale.

It has been purposely shown with two different Y-scales due to the differences in peak amplitudes in the presence of glutamate or kainate.

(18) Fig. 3 sup.1A: Based on the figure legend, horizontal bars representing the application of glutamate are not consistent with time scale bars. Please, check. In the same figure, panel B, the representative traces shown for GluK-1a-Neto1 are not consistent with IK/IG ratio shown in Fig. 3D.

Thanks, we have corrected the horizontal bars representing glutamate application. The representative traces shown for GluK-1a-Neto1 were rechecked and are consistent with the IK/IG ratio shown in Fig. 3D.

(19) I wonder if the authors could discuss the lack of Neto1 effect on the wild type Gluk1-2a channel, as proposed previously.

Sheng et al., 2015 showed that Neto1 enhances the desensitization onset of GluK1. However, it is unclear which GluK1 splice variants were used in that study. GluK1 has several splice variants, but in the present study, we specifically compared GluK1-1a and 2a. In our case, we did not observe the effect of Neto1 on wild-type GluK1-2a in either of the two techniques (whole cell and outside-out patch) we utilized for our study. However, as can be observed from our data, the GluK1-2a receptor alone shows a faster desensitization kinetics than the previous study (Copits et al., 2011). The differences could stem from different experimental conditions such as constructs, recording conditions used etc.

Copits BA, Robbins JS, Frausto S, Swanson GT. Synaptic targeting and functional modulation of GluK1 kainate receptors by the auxiliary neuropilin and tolloid-like (NETO) proteins. *Journal of Neuroscience*. 2011 May 18;31(20):7334-40.

Sheng N, Shi YS, Lomash RM, Roche KW, Nicoll RA. Neto auxiliary proteins control both the trafficking and biophysical properties of the kainate receptor GluK1. *Elife*. 2015 Dec 31;4:e11682. doi: 10.7554/eLife.11682. PMID: 26720915; PMCID: PMC4749551.

<https://doi.org/10.7554/eLife.89755.2.sa0>